THE LOCAL WHITTAKER FACTOR AT A SPLIT LEVEL PRIME, AND AN INCOMPLETENESS IN THE STANDARD RECIPE FOR SCHOFER'S κ

Abstract. Schofer's formula expresses averages of $\log\|\Psi_F\|$ over CM points on a Shimura curve in terms of quantities $\kappa_\eta(m)$ built from derivatives of Fourier coefficients of an incoherent Eisenstein series. The recipe used in practice descends from Kudla–Rapoport–Yang through Schofer and Errthum to Yang, and its central step enumerates the configurations that contribute to the derivative at the centre. That enumeration is derived under a restriction which Kudla–Rapoport–Yang state explicitly — that the Eisenstein series be built from sections given by the characteristic functions of the completions of $\mathcal{O}_k$ — and which fails for the lattices arising from an Eichler order of squarefree level N at a *split* level prime.

We compute the local Whittaker factor of the N-scaled binary lattice at the level prime in closed form for every valuation; its value at the centre is $(N-1)\operatorname{ord}_N(m)$. Re-running the case analysis with this factor in place produces no additional contributing configuration, so no $\log N$ can arise from $\kappa_\eta(m)$ with $m > 0$. The CM values nevertheless require a correction $A_m \log N$, which we measure exactly and check against ground truth on bases with $N \in \{2, 3, 5\}$, one of them forced by a consistency relation independent of any Eisenstein computation.

We then determine that correction outright. Its Eisenstein part is an explicit alternating sum of Gross genus theta series indexed by the subsets of the primes of N at which the tracked isotropic coset is supported — genus sums where the discriminant admits one, telescoped mass-weighted pairs where it does not — valid for every squarefree D and N and every isotropic coset, and independent of which prime of D is split off. A residue identity on $X_0(M)$ makes the correction the pairing of the principal parts of the input form against the Fourier coefficients of that combination, independently of the representative chosen. For D a product of two primes and N prime or 1 — the range of the model set — those coefficients are, in closed form, a Hurwitz class number times one elementary factor per prime of DN, with conductor corrections read off the fundamental decomposition $-4m = D_0c^2$; the rule that the correction vanishes unless $\mathbb{Q}(\sqrt{-m})$ embeds in B_D is part of the formula rather than a side condition, and the empirical rule that it is supported on $N \mid m$ turns out to describe a chosen representative rather than the correction. Twenty bases had suggested a "re-split rule" selecting one splitting of D; there is none to find.

Finally we locate where the transmitted recipe loses the term. The space of weight-1/2 forms for the Weil representation vanishes, so the input form is determined by its principal part and its constant term at the zero coset is genuinely zero; and no per-form normalisation can produce the observed failure, since the uncorrected values are mutually inconsistent on a base where a uniform rescaling could not make them so. What remains is that the $m = 0$ term collapses a sum over isotropic cosets to the zero coset by a condition constraining only the negative-definite part of the coset: the surviving nonzero cosets carry the whole correction. We then evaluate the coefficient they carry, obtaining $\kappa_\eta(0) = -\log N/(N-1)$ at each of the $2N-2$ of them, and summing recovers the measured correction exactly, the factor $\frac{1}{2}$ appearing as $\frac{1}{4} \cdot \frac{2N-2}{N-1}$ and hence independently of N. The $\log N$ arises at the level prime, not at a place of Diff — not because the local factor there vanishes, but because at the zero coset it has a pole, which the passage to a nonzero coset converts into the zero whose derivative is taken. That mechanism is invisible to a case analysis tracking orders of vanishing, which is why the term was lost.

Date: September 5, 2026.

1. Introduction

Let B be an indefinite quaternion algebra over $\mathbb{Q}$ of reduced discriminant D, let $\mathcal{O} \subset B$ be an Eichler order of squarefree level N with $(N, D) = 1$, and let $L = \{\alpha \in \mathcal{O} : \operatorname{tr}\alpha = 0\}$ be the associated lattice of signature $(1, 2)$, so that $L^\vee/L \cong \mathbb{Z}/2 \times \mathbb{Z}/DN \times \mathbb{Z}/DN$ has order $2(DN)^2$ [4, Lemma 8]. For a weakly holomorphic vector-valued modular form F of weight $1/2$ and type ρ_L, Borcherds' lift [1] Ψ_F is a meromorphic modular form on $X_0^D(N)/W_{D,N}$, and Schofer's formula [8, 4] evaluates

$$\sum_{\tau\in \mathrm{CM}(d)} \log|\Psi_F(\tau)| \;=\; -\frac{|\mathrm{CM}(d)|}{4} \sum_{\eta\in L^\vee/L} \sum_{m\geq 0} c_\eta(-m)\,\kappa_\eta(m),$$

valid under the hypotheses $O^+_{L,F} = O^+_L$, $c_0(0) = 0$ and $c_\eta(m) \in \mathbb{Z}$ for $m \leq 0$.

The quantities $\kappa_\eta(m)$ are not given in [4]; the reader is referred to [3] and [13] for their evaluation. Both go back to the computation of Kudla–Rapoport–Yang [5], whose Proposition 2.8 identifies the configurations contributing to the derivative of an incoherent Eisenstein series at the centre and asserts that *these are the only cases which contribute.*

What this note contains. Section 2 traces that assertion to its stated scope. Section 3 computes the missing local ingredient: the local Whittaker factor of the N-scaled binary lattice at a split level prime, in closed form (Theorem 3.1), with the corollary that the empirical "level support rule" of the computational literature is exactly the vanishing of that factor (Corollary 3.2) — a rule which, as emerges later, governs a convenient representative of the correction rather than the correction itself (Remark 10.6). Section 4 re-runs the case analysis with this factor and finds no new case (Proposition 4.1). Section 5 records the correction that the CM values require, computed exactly, and Section 6 lists the explanations we have eliminated.

Section 7 then identifies what the correction reads off — the constant term of the input form at a nonzero isotropic coset — and reduces it, along a route we do not finally take, to seed-0 Eisenstein coefficients of a smaller module; the route stalls on a constant-term identity that the naive weight-3/2 recipe violates, and Proposition 7.1 shows the natural repair is unavailable, since $M_{1/2}(\rho_A) = 0$ for the modules occurring here. Section 8 replaces the Fourier extraction by a finite exact evaluation of that constant term, which reaches bases the numerical oracle cannot and turns up a second, square-class channel in the correction; it ends by locating the aliasing that channel is subject to, in the relation ideal of the space of input forms. Section 9 records the measured derivation of the correction's Eisenstein part: on nine bases, each preregistered against the last, it is a fixed Siegel–Weil combination of Gross genus averages whose apparent base-dependence (the "s-law") is entirely mass normalisation. Six further bases, all preregistered, then test the law where it could break: at composite s the support re-splits (and the three readings we had recorded in advance are all refuted as stated), while the doubling of the weights that accompanies the re-split turns out to be no law at all but the choice of isotropic coset at which the functional is read (Remark 9.1). Four more, at odd level, then separate the last confound and show the $N = 1$ weights following s after all. The twentieth base, $X_0^{1155}(1)$, was computed to adjudicate between the candidate re-split rules and instead removed the question: testing *every* candidate support, rather than the single competitor each earlier base had been given, shows that the rank test separates a one-prime D' from a three-prime one and never selects among the composites (Observation 9.9). Theorem 9.10 then derives this: the two empirical laws are one identity, $\rho = [\Theta(D', s) - \Theta(D', 1)]/[\mathrm{mass}(D', s) - \mathrm{mass}(D', 1)]$, whose two sides are independent of D' because a local computation gives $g_p = \frac{1}{2} f_p$ at every prime and the mass difference telescopes. So the "re-split rule" was never a rule: the competing readings name one object. What survives, now with a proof, is the identity itself, of which the mass-ratio law is the shadow in normalised coordinates.

Section 10 determines the correction rather than merely locating it. Theorem 10.1 generalises the identity to every level and every isotropic tracked coset, as an alternating sum over the subsets of the primes of N at which the coset is supported, with Theorem 9.10 the empty-subset case; Lemma 10.2 supplies the level and multiplier that make its terms usable. Theorem 10.3 is a residue identity on $X_0(M)$ — summing over cosets rather than cusps, which supplies the width factor automatically — from which the correction is the pairing of the input form's principal parts against the Fourier coefficients of that combination (Corollary 10.4), independently of which representative is used; the same identity, applied to a difference of representatives, is why the choice cannot matter. Proposition 10.5 then evaluates those coefficients in closed form, and Remark 10.6 settles how they stand to the fitted weights of Table 1: the two agree on every measured multiplier and disagree at almost every index, because the panels determine the functional only modulo a relation ideal — so the level support rule, and the square-class channel of Section 8 with it, are properties of a chosen representative rather than of the correction. Section 11 takes up the provenance of the correction: of the two readings previously on offer, both can now be excluded, and what survives is that terms inside Schofer's formula were dropped. Those terms are then evaluated. Lemma 11.1 supplies the one missing local ingredient, the $m = 0$ Whittaker factor of a p-scaled plane, by calibrating a normalisation against [9, Lem. 2.18] itself; Proposition 11.2 gives $\kappa_\eta(0) = -\log N/(N-1)$ and recovers the measured correction, factor $\frac{1}{2}$ included.

All numerical statements below are exact rational computations, carried out in Magma; the supporting scripts are described in Appendix A.

2. The chain, and the scope of its central step

Write $U = \lambda^\perp$ for the negative definite plane attached to a CM point of discriminant d, $L_+ = L \cap \mathbb{Q}\lambda$ and $L_- = L \cap U$. Following [13, (11),(12)], one sets

$$\kappa_\eta(m) = \sum_{\mu \in L/(L_+ + L_-)} \sum_{x \in \eta_+ + \mu_+ + L_+} \kappa^-_{\eta_- + \mu_-}\big(m - \langle x, x\rangle/2\big),$$

where $\kappa^-_\mu(m)$ is the s-linear coefficient of the m-th Fourier coefficient $A_\mu(s, m, v)$ of an incoherent Eisenstein series of weight 1 attached to L_-, and $\kappa^-_\mu(0) = 0$ unless $\mu = 0$.

For $m > 0$ one has, with $S_{m,\mu} = \{p : p \mid \Delta \text{ or } v_p(m) > 0\}$ and $\Delta = \operatorname{disc}(L_-)$,

$$A_\mu(s, m, v) = -\frac{2\pi}{L(s+1, \chi_d)} \prod_{p \in S_{m,\mu}} \frac{W_{m,p}(s, \varphi_{\mu,p})}{1 - \chi_d(p) p^{-1-s}}, \tag{1}$$

a genuine Euler product because $\varphi_\mu = \operatorname{char}(\mu + \widehat{L_-})$ is a pure tensor. Since $A_\mu(s, m, v)$ vanishes at $s = 0$, some factor must vanish there, and differentiating (1) yields [13, Lemma 16]: the derivative is the single term at the vanishing place, and vanishes outright if two or more factors vanish.

That step is correct as an identity. What it inherits from [5] is the identification of the vanishing of a local factor with a *representability* obstruction, i.e. with membership of $\operatorname{Diff}(m)$. Kudla–Rapoport–Yang state the scope of their computation plainly:

> “in section 2, we have considered only those Eisenstein series built from sections which are defined by the characteristic functions of the completions of $\mathcal{O}_k$. More general sections could be considered at the cost of (i) more elaborate calculations of Whittaker functions and their derivatives used in computing the

> Fourier expansion of the central derivative, and (ii) incorporation of a level structure in the moduli problem." [5, §6]

(They also fix a maximal order $\mathcal{O}_B$.) The lattice L_- arising here is not of that shape. Write $\Delta = \operatorname{disc}(L_-)$. For a discriminant d with $N \nmid d$ — the case in which the level prime is *unramified* in $k = \mathbb{Q}(\sqrt{d})$, and hence, by Eichler's condition on optimal embeddings, necessarily *split* — one has $\operatorname{ord}_N(\Delta) = 2$: the binary lattice is N-scaled at the level prime, not unimodular. This is exactly a "more general section".

Remark 2.1. The distinction is invisible in the maximal-order setting: there L_- is unimodular at every split prime, no split prime can obstruct representability, and correspondingly the logarithms produced by [8, Theorem 4.1] run over ramified and inert places only. Every discriminant admitting a CM point on $X_0^D(N)$ with $N \nmid d$ has N split, so this configuration is precisely the one the maximal-order theory never meets.

3. The local Whittaker factor at a split level prime

Part (i) of the programme quoted above, at the level prime, is the following computation. Let $X = N^{-s}$.

Theorem 3.1. *Let $d < 0$ be a discriminant with $N \nmid d$ and let L_- be the negative definite binary lattice attached to a CM point of discriminant d, so $\operatorname{ord}_N(\operatorname{disc} L_-) = 2$. Then for every $m \geq 1$, with $j = \operatorname{ord}_N(m)$,*

$$W_{m,N}(X) \;=\; 1 + (N-1)\bigl(X + X^2 + \cdots + X^j\bigr) - X^{\,j+1}.$$

In particular

$$W_{m,N}(1) = (N-1)\operatorname{ord}_N(m), \qquad W'_{m,N}(1) = (j+1)\Bigl(\tfrac{(N-1)j}{2} - 1\Bigr).$$

The statement is a computation rather than a proof: it has been verified in exact arithmetic for $N = 2, 3, 5$ and all $1 \leq m \leq 60$, on the lattices L_- of $X_0^{15}(2)$, $X_0^{6}(5)$ and $X_0^{10}(3)$ at firing discriminants, giving 180 independent checks. It is pinned as a regression test in the accompanying code. That test alone would establish only that the code is self-consistent, since the closed form was read off the same implementation; the value $W_{m,N}(1) = (N-1)\operatorname{ord}_N(m)$ is recovered independently at the end of §11, from brute-force local representation densities computed by counting.

Corollary 3.2. $W_{m,N}(1) = 0$ *if and only if* $N \nmid m$.

This identifies, and explains, an empirical rule that has been observed repeatedly in computations with these curves: the correction discussed in Section 5 admits a representative supported exactly on the indices m with $N \mid m$. That rule is the vanishing of the level-prime factor. It is a statement about a representative rather than about the correction itself — the coefficients determined in closed form in Proposition 10.5 are supported by an embedding condition at the primes of D, and are nonzero at many m with $N \nmid m$. Why the two are nevertheless compatible, and why no panel of the kind used below can tell them apart, is Remark 10.6.

Remark 3.3. At $j = 0$ the factor is $W_{m,N}(X) = 1 - X = 1 - N^{-s}$, the inverse local zeta factor. Its vanishing at $s = 0$ is therefore of a different nature from a representability obstruction: it holds identically for all m with $N \nmid m$, and at a split N the local quadratic space represents every m.

Remark 3.4. The value $W_{m,N}(1) = (N-1)\,\mathrm{ord}_N(m)$ coincides with the local factor Kudla–Yang obtain at $p \mid N$ for the *quaternary* Eichler lattice [6, (9.1)]. The agreement appears to be a genuine coincidence of shape between the quaternary lattice and the binary L_-; we do not have an explanation for it.

4. Re-running the case analysis

With Theorem 3.1 in hand one can redo the enumeration of [5, Prop. 2.8] for the N-scaled section. Normalising as in (1), put $f_p(s) = W_{m,p}(s)/(1-\chi_d(p)p^{-1-s})$.

Proposition 4.1. *At a firing discriminant, f_N has a simple zero at $s=0$ exactly when $N \nmid m$, and this zero is not a* Diff *zero. Consequently*

$$\mathrm{ord}_{s=0}\, A_\mu(s,m,v) \;=\; |\mathrm{Diff}(m)| + [\, N \nmid m\,],$$

and since incoherence forces $|\mathrm{Diff}(m)| \geq 1$:

1. *if $N \mid m$, the order is $|\mathrm{Diff}(m)|$, and the derivative is nonzero precisely when $|\mathrm{Diff}(m)| = 1$, contributing a logarithm at the ramified or inert place of* $\mathrm{Diff}(m)$*;*
2. *if $N \nmid m$, the order is at least 2 and the derivative vanishes.*

In neither case does a $\log N$ occur.

So the level structure only ever raises the order of vanishing: it removes contributions, and creates no new contributing configuration. The three cases of [5, Prop. 2.8] stand, with no fourth. This is confirmed computationally: over roughly 190 terms (μ, x) arising in the evaluation of $\kappa_0(m)$ on three bases, the level prime is *never* the unique vanishing place, while it lies among two or more in 60–80% of terms.

Corollary 4.2. *Within this framework, a $\log N$ can enter a CM value only through the $m = 0$ term $c_0(0)\kappa_0(0)$, where $\kappa_0(0)$ carries the summand $\sum_{p\mid N/(N,d)} \log p$ of* [13, Remark 15(2), Lemma 20]*.*

"Within this framework" is the operative clause: the term the framework drops also carries a $\log N$, and from the same prime, but at the *nonzero* isotropic cosets rather than at 0 (Proposition 11.2). Since $c_0(0) = 0$ in our situation, the term retained here vanishes and the dropped ones carry the whole correction.

5. The correction the CM values require

Against this stands a computation of the input form itself. Following [4, eq. (6)] one may form $F_f = \sum_{\gamma\in\Gamma_0(M)\backslash SL_2(\mathbb{Z})} (f|_{1/2}\gamma)\,\rho(\gamma^{-1})e_0$ and evaluate it numerically; the metaplectic phase is handled by evaluating both the slash and ρ along one and the same ST-word, so that they share a single metaplectic lift. Every such evaluation is gated on reproducing the principal part of F_f predicted by [4, Lemma 24] from the two scalar q-expansions of f; on the runs reported here that gate is met to 10^{-103}.

Observation 5.1. For every Borcherds form produced by the pipeline on $X_0^{15}(2)$, $X_0^{6}(5)$, $X_0^{10}(3)$ and $X_0^{21}(2)$ one finds $c_0(0) = 0$ (to 10^{-103}), so the hypothesis $c_0(0) = 0$ of [4, Thm. B] is satisfied and, by Corollary 4.2, no $\log N$ can arise.

Observation 5.2. The isotropic cosets of $L^\vee/L$ number exactly $2N-1$; the D-part of the discriminant form is anisotropic, so all of them come from the level-N hyperbolic plane. All $2N-2$ nonzero isotropic cosets carry the same constant term $c_\eta(0)$.

Both counts in Observation 5.2 are counts for *prime* N, which is the only case that occurs here — every base in this project, and every entry of the model set, has N prime or $N = 1$. The agreement of the nonzero cosets, recorded here as an observation, becomes a theorem in §10, where the general-N counts are given; it fails as soon as N is composite.

Observation 5.3. Let $\mathrm{mult}(f) = \frac{1}{2}c_\eta(0)$ for any nonzero isotropic η. Then adding $\mathrm{mult}(f) \cdot \sum_{p|N/(N,d_{\mathrm{fund}})} \log p$ to the computed CM value reproduces, exactly, the corrections independently measured by kernel-consistency on $X_0^{15}(2)$ (9 values), $X_0^{6}(5)$ (5) and $X_0^{10}(3)$ (5), and by the consistency relation of Observation 5.4 below on $X_0^{14}(5)$ (1, at eleven digits): 20 of 20. A further base $X_0^{21}(2)$ supplies nine values with no prior ground truth against which to check them. Ground truth here has to be produced rather than looked up: a published CM value cannot in general test the correction, for the reason set out in Remark 11.3.

That the correction is genuinely required, and not an artefact of a normalisation, is visible on $X_0^{6}(19)$: on a set of CM points including discriminants divisible by N, the uncorrected values are mutually *inconsistent* — the associated system has no solution — whereas with the correction the published model is recovered.

Ground truth without an Eisenstein computation. The pipeline imposes, at every rational CM point d, the consistency relation

$$\varepsilon_1(d)\,\frac{s(d)}{s(d_A)} + \varepsilon_2(d)\,\frac{\tilde{s}(d)}{\tilde{s}(d_B)} = 1, \qquad \varepsilon_i(d) \in \{\pm 1\}, \tag{2}$$

between the two Hauptmodul rows, normalised at the discriminants d_A, d_B where they vanish. Restoring a correction $r_f \log N$ to $\log|\Psi_f|$ multiplies the normalised value at d by $N^{\,r_f(k(d)-k(d_{\mathrm{ref}}))}$, where $k(d) \in \{0, 1\}$ records whether the term fires at d. Each rational CM point whose firing status differs from the reference's therefore contributes one equation in (N^{r_1}, N^{r_2}); two determine the pair and any further points are consistency checks. This measures the multipliers of the two Hauptmodul forms on any base the pipeline reaches, in seconds, with no Eisenstein series evaluated.

Observation 5.4. On nine bases the relation (2) determines the pair (r_{-1}, r_{-2}) uniquely:

$X_0^{15}(2)\colon (4,2)$	$X_0^{14}(5)\colon (1,1)$	$X_0^{6}(7)\colon (1,2)$	$X_0^{6}(5)\colon (0,0)$
$X_0^{10}(3)\colon (0,0)$	$X_0^{10}(11)\colon (0,0)$	$X_0^{6}(19)\colon (0,0)$	$X_0^{22}(7)\colon (0,0)$
$X_0^{34}(5)\colon (0,0)$			

Every value with an independent determination is reproduced — $X_0^{15}(2)$ against the kernel-consistency measurement, $X_0^{6}(5)$ and $X_0^{10}(3)$ against the numerical evaluation of F_f, and $X_0^{6}(7)$, the one *nonzero* cross-check, against an earlier independent measurement. Conversely, on $X_0^{14}(5)$ the pair $(1,1)$ was measured first by (2) and *then* confirmed by the numerical evaluation of F_f: $\frac{1}{2}c_\eta(0) = 1.00000000000$ on the form with pole order 16, with the principal-part gate met to $2.3 \cdot 10^{-37}$ and all eight nonzero isotropic cosets agreeing. This is the first verification of the rule of Observation 5.3 at $N > 2$ against ground truth obtained by an entirely independent method.

Writing $\mathrm{mult}(f) = \sum_m c(-m)A_m$ and using Borcherds' obstruction to set $A_m = -b(m)/4$ with b the relevant weight 3/2 Eisenstein coefficient, the linear functional is determined per base by the measured multipliers. Without further hypotheses the system leaves four free directions on each base; imposing the support rule of Corollary 3.2 (the weight vanishes unless $N \mid m$) removes them all, and the resulting weights are then unique, with six spare conditions satisfied on each of the three bases. Table 1 records the result. Those spare conditions say that the $N \mid m$ ansatz *lies inside* the solution space, not that it is forced: the underlying system is underdetermined, and this is a choice of representative within it. Remark 10.6 makes that precise once the coefficients are known in closed form, and exhibits a second representative — the true one — agreeing with Table 1 on every measured multiplier while differing from it at almost every index.

base	A_m (equivalently $b(m) = -4A_m$)		
$X_0^{15}(2)$	$A_2 = -1$	$A_{10} = 1$	$A_{30} = 0$
$X_0^{6}(5)$	$A_{10} = 3/2$	$A_{15} = 1/2$	$A_{30} = 3/2$
$X_0^{10}(3)$	$A_3 = 1/2$	$A_{12} = 1/2$	$A_{30} = 3/2$
$X_0^{21}(2)$	$A_2 = -2$, $A_6 = -4$	$A_{18} = 2$, $A_{42} = -8$	(0-block: $B_{21} = -5$)

Table 1. The correction, per index, as a linear functional of the principal part, in the representative fixed by the support rule. The values are d-independent, and the recipe of §2 returns 0 where the table does not. The individual A_m are determined only modulo the relation ideal of the panel; see Remark 10.6.

6. Explanations eliminated

Each of the following was tested against the data of Section 5 and is inconsistent with it. We record them because several are natural first guesses.

1. mult $= c_0(0)$, or any linear combination of the two scalar constant terms $\mathrm{Coeff}(f_\infty, 0)$ and $\mathrm{Coeff}(f_0, 0)$: inconsistent on three bases.
2. Evaluating the local Whittaker at a nonzero isotropic coset η^* rather than at 0: destroys the pattern of vanishing indices on all three bases. The mechanism is now clear: the nonzero isotropic cosets live in the level-N hyperbolic plane, so changing the coset changes *only* the level-prime factor (measured: from $1 - X^2/N$ to the constant 1, with the factors at $p \mid D$ byte-identical) and can never repair an index whose defect is at a ramified prime. This is not in tension with the conclusion of §11, that the correction is carried by the $m = 0$ term at precisely those cosets: what is eliminated here is moving the coset inside the $m > 0$ Whittaker factors, which is a different object from the $m = 0$ constant term $c_{\eta^*}(0)$ that §7 onwards reads off.
3. Replacing the local factor at $p \mid D$ by the anisotropic branch of [6, Prop. 5.3]: does not reproduce b, under either sign of κm. (The implementation of [6, Props. 5.3, 5.4] agrees with the pipeline's own factor at every good prime, 470 checks, and at the level prime, 150 checks.)
4. The incoherent support condition $|\mathrm{Diff}(m)| = 1$: scores 8/9 on the exact coefficients and is the only rule that predicts $b(30) = 0$ on $X_0^{15}(2)$, but is far too permissive — for $r \leq 60$ it holds at 15, 19 and 20 indices with $N \nmid r$, where the coefficient must vanish.
5. Cancelling the zeta factor $1 - X$ at the level prime before counting vanishing places: the recovered terms differentiate at ramified and inert primes, never at N.
6. The inner $m = 0$ of κ_η, i.e. terms with $\langle x, x\rangle/2 = m$: never occurs, in all 60 (discriminant, index) pairs over every firing discriminant of the three bases.
7. Schofer's outer $m = 0$ term: vanishes by Observation 5.1.
8. The re-enumeration of §4: adds no case.
9. Any re-weighting of the existing local data: every combination of a subset of the prefactors $\{\sqrt{m}/\sqrt{|d|}$, h/w, $\prod(1 - \chi(p)/p)$, $\prod(1 - 1/p^2)^{-1}\}$ with a local rule drawn

from {the product of the $W_p(1)$; the derivative of that product; the derivative taken at N, at the primes $p \mid D$, at each single prime, or at the vanishing places}, times an arbitrary constant per base, fails to reproduce the nine measured multipliers of $X_0^{15}(2)$.

(10) More strongly, *no multiplicative correction of any kind* can succeed: at two of the nine determined indices (both with $m = 10$) the product of the code's local factors vanishes while the true weight does not, so the discrepancy is infinite there. The missing contribution is a *term*, not a factor.

(11) Differentiating at the place where the ternary *space* fails to represent m (Kudla's Diff, computed at the level of quadratic spaces rather than lattice densities): reproduces $A_2 = -1$ and $A_{30} = 0$ on $X_0^{15}(2)$ exactly — the only rule found that *explains* $A_{30} = 0$ — but fails on 7 of 9 indices, and is refuted as a support rule by $X_0^6(5)$ at $m = 15$, where Diff is empty but $A_{15} = 1/2 \neq 0$. The support is $N \mid m$; whether Diff supplies the correct place to differentiate is a separate question.

7. The correction identified

This section identifies the object the correction reads off, and reduces it — along one route — to seed-0 Eisenstein coefficients of a smaller module. That route is not the one finally taken: §10 determines the correction outright by a different argument, and the reader interested only in the answer may pass to it. Two things here are used later regardless. Proposition 7.1, that $M_{1/2}(\rho_A) = 0$, is what closes the first of the two readings in §11; and the failure recorded at the end of this section is what sent the derivation to the exact evaluation of §8.

The rule of Observation 5.3 reads off the constant term $c_{\eta^*}(0)$ of the input form at a nonzero isotropic coset η^*. By the constant-term pairing (pair F against a holomorphic Eisenstein series of the complementary weight 3/2 and take the constant term of the resulting weight-2 invariant), that quantity is determined by the principal part of F paired against the coefficients of E_{A,η^*}, the Eisenstein series attached to the coset η^* of the discriminant form $A = L^\vee/L$. This is precisely the object the maximal-order literature does not compute: [2], [6, §8–9] and the recipe of §2 all work with the series attached to the *zero* coset (equivalently, with sections $\operatorname{char}(\mu + \widehat{L})$ seeded at 0), and Kudla–Yang explicitly leave the general Eichler-order case open at the end of their §9.

The coefficients of $E_{A,\beta}$ for isotropic $\beta \neq 0$ are computed by Schwagenscheidt [10]. For our purposes the essential ingredient is not the main formula but the oldform relation [10, Prop. 1.4] applied to the trivial character: for $\beta = \eta^*$ of prime order $N_\beta = N$,

$$E_{A,\eta^*,\chi_0} \;=\; E_{B,0}\!\uparrow_B^A \;-\; E_{A,0}, \qquad B = \langle \eta^* \rangle^\perp / \langle \eta^* \rangle \;\cong\; L_1^\vee / L_1, \tag{3}$$

where L_1 is the even lattice generated by L and η^*. Pairing F against (3) and using Observations 5.1 and 5.2 — $c_0(0) = 0$, the constant $c_\eta(0)$ shared by all $2N-2$ nonzero isotropic cosets, and the vanishing of the constant-term pairing for every nontrivial character twist — gives

$$\operatorname{mult}(f) \;=\; -\frac{1}{4(N-1)} \sum_{n>0} \sum_{\gamma \in (\eta^*)^\perp} c_\gamma(-n)\, b^{B,0}_{\gamma+H}(n), \qquad H = \langle \eta^* \rangle, \tag{4}$$

in which every object on the right is a *seed*-0 Eisenstein coefficient of the smaller module B — the territory of [2], computable by the same machinery as the existing recipe, applied to L_1 in place of L.

Three structural facts corroborate the identification.

(1) The pairing constant in (4) is $4(N-1)$ — exactly the normalisation measured in Observation 5.3 (the sum of $c_\eta(0)$ over the $2N-2$ nonzero isotropic cosets equals $4(N-1) \cdot \operatorname{mult}$).

(2) $\det L_1 = \det L/N^2$: the level-prime part of L_1 is *unimodular*, so the coefficients $b^{B,0}$ have no vanishing level-prime factor. The zero $W_{m,N}(1) = 0$ at $N \nmid m$ (Corollary 3.2), which removes the majority of terms from the seed-0 pairing on A, never occurs on B — the structural mechanism by which (4) can produce the $\log N$ terms that §4 shows the seed-0 recipe cannot.

(3) $2|\det L_1|$ remains a perfect square (e.g. $2 \cdot 450 = 30^2$ on $X_0^{15}(2)$), so the discriminant and character bookkeeping of the weight-3/2 coefficient formula carries over unchanged in shape.

One caution, which the data has since promoted to a finding. The identity (3) is proven in [10] for weight $k \geq 5/2$; at $k = 3/2$ both sides require analytic continuation, and the naive pairing argument assumes the continued series is holomorphic at the special point, where Zagier-type non-holomorphic contributions are possible. That assumption *fails*, and it fails in a way one can measure without any of the machinery of this section: for the true seed-0 series on A itself, the constant-term pairing forces $\sum_{n>0,\gamma} c_\gamma(-n)\, b_\gamma^{A,0}(n)$ to equal $-c_0(0)$ minus a spread term proportional to the multiplier — so on a form with mult $= 0$ and $c_0(0) = 0$ it must vanish outright. Evaluating the left side with the naïve weight-3/2 coefficient formula (rank 3 collapses the Bruinier–Kuss local polynomials to plain representation densities, since they are evaluated where the factor $1 - p^{m-1}X$ vanishes) gives -5, not 0, on such a form of $X_0^{15}(2)$. The naïve recipe therefore violates its own constant-term identity, independently of everything specific to (4) — and this, rather than any defect of (3), is where the computation of (4) currently stops.

One natural reading of that violation — that the weight-3/2 series is merely harmonic, and the missing term is the pairing against its ξ-image, a weight-1/2 theta on A — is *refuted* by the following observation, which seems worth recording in its own right.

Proposition 7.1. *For the discriminant forms $A = L^\vee/L$ arising here (and for their rescalings $(2L)^\vee/2L$), the space of holomorphic vector-valued modular forms of weight $1/2$ for ρ_A (and for ρ_A^*) is zero.*

Proof sketch. By the Serre–Stark theorem in its vector-valued form [11, 12], every such form is a linear combination of unary theta functions; each arises from a cyclic nondegenerate orthogonal summand Z of a quotient $H^\perp/H$ of A by an isotropic subgroup H, tensored with a ρ_C-invariant vector on the complement C, and invariant vectors are supported on self-dual isotropic subgroups of C. The D-part of A is anisotropic of shape $(\mathbb{Z}/p)^2$ at each odd $p \mid D$ (Observation 5.2); an isotropic subgroup cannot meet it, a cyclic summand cannot absorb both $\mathbb{Z}/p$ factors, and passage to $H^\perp/H$ preserves it. Hence at least one anisotropic $\mathbb{Z}/p$ survives into every complement, no invariant vector exists, and every candidate theta vanishes. □

In particular $E_{A,0}$ of weight 3/2 has no shadow, the completion reading is closed, and the input form F_f is determined by its principal part alone — a fact the computations of §8 confirm to seventy digits on pairs of forms with equal principal parts. What the failed identity signals is therefore not a non-holomorphic correction on A but a missing *channel* in the ansatz for the functional itself, which the exact evaluation of the next section locates.

8. The constant term evaluated exactly, and the two channels of the correction

The constant term that Observation 5.3 reads off admits a finite evaluation which replaces the Fourier extraction of §5 entirely. For a coset word w with matrix g and an eta monomial $\prod_d \eta(d\tau)^{r_d}$ of f, triangularising $\left(\begin{smallmatrix} d & 0 \\ 0 & 1 \end{smallmatrix}\right) g = g_d \begin{pmatrix} a_d & b_d \\ 0 & e_d \end{pmatrix}$ exhibits $(f|_{1/2}w)$ as a constant multiple of a q-series with *exact rational exponents* and coefficients that are roots of unity; the single constant per (monomial, word) is pinned numerically at one point of $\mathbb{H}$ and verified at a second, so no eta-multiplier or metaplectic bookkeeping enters. The coefficient of q^0 in $f|_{1/2}w$ is then a finite sum, and

$$c_{\eta^*}(0) \;=\; \sum_{w} \big(\rho(w^{-1})e_0\big)_{\eta^*}\, a_0(f|_{1/2}w)$$

is evaluated in minutes per base, with no horocycle sampling, no aliasing, and none of the cancellation constraints of §A. The same series supply the full principal part of F_f at every coset and exponent; on $X_0^{22}(3)$ this reproduces [4, Lemma 24] coefficient by coefficient to 10^{-70}, and on $X_0^{15}(2)$ the nine multipliers of Observation 5.3 are reproduced exactly, the two zero multipliers to 10^{-71}.

Two panels beyond the reach of the numerical oracle result. On $X_0^{22}(3)$ the nine multipliers are

$$(r_{-2}, r_{-1}, r_9, \ldots, r_{15}) = (3,\ 3,\ -6,\ 3,\ -3,\ -9,\ 3,\ 6,\ -6),$$

and on $X_0^{21}(2)$ — a base on which no ground truth of any kind existed —

$$(2,\ 2,\ 4,\ 2,\ 2,\ 2,\ -4,\ -4,\ 0).$$

The second channel. The nine $X_0^{22}(3)$ values are *inconsistent* with any linear functional of the principal part supported on the indices $N \mid m$: the combinations of forms $(f_{-2} - f_9 + f_{11})$ and $(-f_{-1} - f_9 + f_{12})$ cancel every N-divisible index of the principal parts identically, yet their multipliers sum to 6 and -6. The residual support of both combinations is the set $m \in \{1, 4, 16, 25\}$ — perfect squares — with residual coefficients $(4, 4, 2, 2)$ and $(-4, -4, -2, -2)$, so both reduce to the single equation $12\, w_\Box = 6$. The correction therefore carries a second channel:

$$\mathrm{mult}(f) \;=\; \underbrace{\sum_{N|m} c_0(-m)\, w(m)}_{\text{level channel}} \;+ w_\Box(D,N) \sum_{k\geq 1} c_0(-k^2), \tag{5}$$

with $w(m)$ depending on m only through the square class of m/N. With both channels present, the measured multipliers of *every* base are consistent, with spare conditions throughout; the data force

$$w_\Box = 0 \text{ on } X_0^{15}(2),\ X_0^{6}(5),\ X_0^{6}(13),\ X_0^{6}(17); \qquad w_\Box = \tfrac{1}{2} \text{ on } X_0^{22}(3); \qquad w_\Box = 1 \text{ on } X_0^{21}(2).$$

The square-class sum is the signature of the *scalar* Zagier weight-3/2 structure — which is consistent with Proposition 7.1: no vector-valued shadow exists on A, but the scalar theta enters through the cusp expansions of f itself. The bases with $w_\Box = 0$ are exactly those on which the level-support rule appeared exact, which is why this channel escaped notice through eighteen bases of fitted corrections.

Conjecture 8.1. *$w_\Box(D,N) \neq 0$ if and only if no prime of DN splits in $\mathbb{Q}(i)$ — the embedding criterion for the orders of discriminant $-4k^2$ — and in that case $w_\Box = 1/(N-1)$.*

The condition matches all six determined values above (DN contains a prime $\equiv 1 \bmod 4$ precisely on the four bases with $w_\Box = 0$), and the magnitude matches $N = 2, 3$. We record, in advance of the measurements, the values the conjecture predicts on the panels currently being computed: 1 on $X_0^{33}(2)$; $\frac{1}{2}$ on $X_0^{38}(3)$ and $X_0^{14}(3)$; 0 on $X_0^{35}(2)$, $X_0^{22}(5)$ and $X_0^{34}(5)$; $\frac{1}{6}$ on $X_0^{22}(7)$; $\frac{1}{10}$ on $X_0^{6}(11)$.

Scorecard at the time of writing. The panels for $X_0^{33}(2)$, $X_0^{35}(2)$, $X_0^{22}(5)$, $X_0^{38}(3)$, $X_0^{14}(3)$, $X_0^{6}(11)$, $X_0^{10}(11)$, $X_0^{14}(5)$, $X_0^{6}(19)$, $X_0^{34}(5)$ and $X_0^{22}(7)$ have since been measured. Every one is consistent with the two-channel form (5), with spare conditions; two of the preregistered predictions have been decided, and both *hit*: $X_0^{14}(5)$ pins $w_\square = 0$, and $X_0^{34}(5)$ pins $w_\square = 0$ at rank 7 of 9 (both have $5 \mid DN$ split in $\mathbb{Q}(i)$); the latter panel also pins the k-twist values $w(1,1) = w(1,2) = w(1,4) = 0$, $w(17,1) = 0$ and $w(2,1) = w(34,1) = \frac{1}{2}$. On the remaining bases the nine-form panel leaves $w_\square$ undetermined: the square-class column of the principal-part profile matrix lies in the span of the level-channel columns, so no value of $w_\square$ is forced either way. This aliasing is *intrinsic* rather than an accident of the panel — see the assembly analysis below — so the outstanding predictions cannot be decided by further panels of the same kind. Nor, as it turns out, by the derivation. Section 10 determines the functional outright, and its canonical representative — the Fourier coefficients of an explicit weight-$\frac{3}{2}$ form — does not have the two-channel shape of (5) at all. So $w_\square$ is a coordinate of a gauge-fixed representative rather than an invariant of the correction, and Conjecture 8.1 is to be read as a statement about that representative (Remark 10.6). Its predictions stand unfalsified where they have been tested; what changes is that no computation can be expected to settle the rest.

The level channel. Within the first channel the data pin two further structural facts. The weight at $m = Ndk^2$ (d squarefree) depends on k through a sign: on $X_0^{10}(7)$ the two Hauptmodul rows force $w(7) = +\frac{1}{2}$ and $w(28) = -\frac{1}{2}$, and across all bases the k-dependence is consistent with the twist $\lambda(k)\left(\frac{k}{N}\right)$ (Liouville times Legendre), while k-independence and k-linearity are refuted. And no assignment of the weights by a fixed product of Legendre symbols in (D, N, d) covers all bases: three successive such laws, each consistent with all data available when it was formulated, were refuted by the next base measured ($X_0^6(13)$, then $X_0^{10}(7)$, then $X_0^{22}(3)$ and $X_0^{38}(3)$). The surviving magnitudes $0, \pm\frac{1}{2}, \pm 1, \pm\frac{3}{2}$ vary with the base in the manner of representation numbers rather than characters. Since the coset sum above evaluates $c_{\eta^*}(0)$ as a finite sum of Weil-representation entries — quadratic Gauss sums — against the cusp expansions of f, grouping it by cusp class is a finite, exact route to the closed form; we have executed it, with the following results.

The cusp-class assembly. Throughout this subsection $A = L^\vee/L$ carries the *negated* quadratic form $\bar{q} = -q$ (the dual Weil representation of the evaluation above is the representation $\rho_{\bar{A}}$ of [7] attached to $\bar{A} = (A, \bar{q})$; the identification is verified coset by coset in `rho5.m`), η^* denotes a nonzero isotropic element, M is the lattice level, and for $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ with lower row (c, d) we write $g = \gcd(c, M)$ for its cusp class. We assume DN even, the case of all computations in this paper.

Proposition 8.2 (local structure)**.** *The Jordan components of $\bar{A}$ are as follows. For each odd $p \mid D$ the p-part is the anisotropic plane $(\mathbb{Z}/p)^2$; for each odd $p \mid N$ it is the hyperbolic plane; the 2-part has order 8 and equals $A_2^1 \oplus C_2$ if $2 \mid N$, and $A_2^1 \oplus B_2$ if $2 \mid D$, in the notation of* [7, §2]. *Consequently the Gauss function of the 2-part on odd arguments is*

$$G_2(x) \;=\; \epsilon_D \left(\tfrac{2}{x}\right) e_8(x), \qquad \epsilon_D = \begin{cases} +1, & 2 \mid N, \\ -1, & 2 \mid D, \end{cases}$$

by [7, Lemmas 3.7, 3.8] *(the B_2 factor contributes the constant $\left(\frac{3}{2}\right) = -1$).*

Proof. The completion $L \otimes \mathbb{Z}_p$ depends, up to isometry, only on the local type of the order: at $p \mid D$ the maximal order of the ramified quaternion algebra over $\mathbb{Q}_p$ is unique up to conjugation, and at $p \mid N$ so is the Eichler order of level p in $M_2(\mathbb{Q}_p)$. Hence the p-part of the discriminant form is a single isomorphism class determined by the case, and one computation per case identifies it: the odd parts are the standard anisotropic (ramified) and hyperbolic (split Eichler) planes, and for $p = 2$ the $\bar{q}$-value multisets $\{0, 0, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$ $(2 \mid N)$ and

$\{0, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{3}{4}, \frac{3}{4}, \frac{3}{4}\}$ $(2 \mid D)$, computed in `g2jordan.m` and constant across ten bases, identify $A_2^1 \oplus C_2$ and $A_2^1 \oplus B_2$ respectively. ☐

Theorem 8.3 (the ρ-entry at a cusp class)**.** *Let γ have lower row (c, d) with class g, and let $\widetilde{\gamma}$ be its canonical metaplectic lift. Then $(\rho_{\bar{A}}(\widetilde{\gamma})e_0)_{\eta^*}$:*

(1) *vanishes iff $N \mid g$ or $c \equiv 2 \pmod 4$;*
(2) *has absolute value $g/\sqrt{|A|}$ for odd g and g/M for $4 \mid g$;*
(3) *for odd g and the canonical representative ($a = d = 1$, $c = -g$) equals $(g/\sqrt{|A|})\, e_8(E_0(g))$ with*

$$E_0(g) \;=\; 2 + 4\,r(g) - 4\,\omega(g) + \arg_8 G_2(-g) \mod 8, \qquad \arg_8 G_2(x) = x + 4\left[\left(\tfrac{2}{x}\right) = -1\right],$$

where $r(g) = \#\{p \mid g : p \equiv 3 \bmod 4\}$ and $\omega(g)$ the number of prime divisors; in particular E_0 is independent of the base (D, N);
(4) *for $4 \mid g$ and the canonical representative it is real, of sign $\mu(g/4)$.*

Proof. By [7, Thm. 6.4], for $x = 0$ the η^*-component of $\rho_{\bar{A}}(\widetilde{\gamma})e_0$ is the *single* term $\xi(a,c)\sqrt{|A[c]|/|A|}\; e(ac\,\bar{q}(y_0) + B(x_c, y_0))$ if $\eta^* \in cA + x_c$ (with $cy_0 + x_c = \eta^*$), and 0 otherwise. (i): for $N \mid c$ the N-part of cA is trivial while η^* has nonzero N-part, so the support condition fails. It also fails when $c \equiv 2 \pmod 4$: by [7, Lem. 2.4] the 2-adic element x_c is nonzero exactly when $2 \,\|\, c$, since A_2 carries an odd Jordan component of scale 2 (Proposition 8.2); then x_c has nonzero 2-part while cA has trivial 2-part, so no element of $cA + x_c$ has trivial 2-part, whereas every nonzero isotropic η^* does (isotropy splits 2-adically, the odd part having odd order). Conversely for $N \nmid g$ and $c \not\equiv 2 \pmod 4$ the support condition holds. (ii): $|A[c]| = \prod_{p \mid g \text{ odd}} p^2$ times 8 if $2 \mid c$, and $|A| = M^2/2$; the 2-part is annihilated already at $2 \mid c$ because $A_2 \cong (\mathbb{Z}/2)^3$ has *exponent* 2, although its level is 4. (iii): choose y_0 inside the N-part; then $\bar{q}(y_0) = c^{-2}\bar{q}(\eta^*) = 0$ and (for odd c) $x_c = 0$, so the phase is exactly $\xi(a,c)$, which [7, Def. 6.1] factors as $e_4(-\text{sign})\,\xi_0\,\xi_2 \prod_J \xi(J)$ over the Jordan components of Proposition 8.2. At the canonical representative: $\xi_2 = 1$; $\xi_0 = (\frac{-1}{-g})(-1, -g)_\infty = e_8(4 + 4r(g))$; each anisotropic pair with $p \nmid g$ contributes $(\frac{t_1 t_2}{p})e_8(2 - 2p) = -1$ and each hyperbolic pair $+1$ ([7, Lem. 3.6], independently of the argument); the components at $p \mid g$ contribute 1; and the 2-part contributes $G_2(-g)$ of Proposition 8.2. Collecting, and using $e_4(-\text{sign}(\bar{A})) = e_8(-2)$ together with the two Gauss-sum factors $e_8(-3)$ and $e_8(1)$ arising from the normalisation of [7, eq. (6.6)] (the latter because the auxiliary $d' \equiv 1 \bmod 8$ forces its Kronecker symbol to $+1$), gives $E_0(g) = 2 + 4r - 4\omega(g) + \arg_8 G_2(-g) + 4\,\omega_{\text{odd}}(D) + 4\,[2 \mid D]$. The last two terms cancel: an indefinite quaternion algebra ramifies at an even number of finite primes, so $\omega_{\text{odd}}(D)$ is odd exactly when $2 \mid D$, which is exactly when $\epsilon_D = -1$ shifts $\arg_8 G_2$ by 4. This proves (iii) and the base-independence. (iv): for $4 \mid c$ the level-4 component is annihilated, $x_c = 0$ again, and every factor of ξ is real ($\xi_2 = e_8(-2(c_2 - 1 + s_2))$ with c_2 the odd part of c and s_2 the 2-part signature contributes a sign, as do ξ_0 and the anisotropic pairs prime to g); the product of signs is $\mu(g/4)$ by the same bookkeeping. ☐

The theorem reproduces all nine measured odd classes ($g = 1, 3, 5, 7, 11, 15, 21, 33, 35$ across six bases) and all four measured $4 \mid g$ classes. On $\omega(g) \le 2$ — every class measurable on these bases — E_0 coincides with the symbol form $1 + 6r(g) + 4s(g)$, $s(g) = \#\{p < q \mid g : (\frac{p}{q})(\frac{q}{p}) = +1\}$, but the two diverge from $\omega(g) = 3$ onward: the symbol form is only the low-complexity shadow of E_0 (`rho5.m`, `thetag-derivation.md`). Word-based evaluations differ from the canonical lift by the metaplectic centre (± 1 per word, the Kubota cocycle product of the letters), which cancels in the pairing below.

Proposition 8.4 (class constancy)**.** *Let f be weakly holomorphic of weight $1/2$ for $\Gamma_0(M)$ whose multiplier restricted to $\Gamma_0(\mathrm{lev}(\bar{A}))$ equals the character χ by which that group acts on e_0 (the case of all Borcherds inputs here, forced by well-definedness of the coset average of* [4, Lem. 24]*). Then $(\rho(w^{-1})e_0)_\eta\, a_0(f|_{1/2}w)$ is, for every isotropic η, constant as w ranges over a cusp class, and consequently*

$$c_{\eta^*}(0) \;=\; \sum_g N_g \left(\rho(w_g^{-1})e_0\right)_{\eta^*} a_0(f|_{1/2}w_g)$$

with one arbitrary coset w_g per class and N_g the number of cosets in the class.

Proof. Replacing w by $\gamma w T^k$ with $\gamma \in \Gamma_0(M)$ multiplies $a_0(f|_{1/2}w)$ by $\chi(\gamma)$ (the T-shift fixes the q^0 coefficient), while e_0 is a $\rho(\Gamma_0(\mathrm{lev}))$-eigenvector with character χ ([7, Lem. 5.5, 5.6]; verified to 10^{-59} in `rho6.m`), so the ρ-side acquires $\overline{\chi(\gamma)}$, and the isotropy of η makes its component T-invariant. The factors cancel. □

The constancy is verified on 570 class/form/base checks (`cuspclass2.py`), and Theorem 8.3 with Proposition 8.4 make the evaluation of $c_{\eta^*}(0)$ a sum over *classes* rather than cosets — the form in which it is deployed in production (`M0MultiplierExact`), with the constancy re-verified at runtime on sampled cosets. The remaining ingredient, $a_0(f|_{1/2}w_g)$, is itself closed-form: the pinned constants of the evaluation equal $\zeta_w \prod_d \left[\varepsilon(g_d)\, e(b_d/24e_d)\, e_d^{-1/2}\right]^{r_d}$ with ε the Dedekind-sum eta multiplier and ζ_w the metaplectic cocycle unit of the word, certified on all 144 cosets of $X_0^{15}(2)$ and all 288 of $X_0^{10}(7)$ (`cusp4.m`), so nothing in the assembly remains numerical.

Because $c_{\eta^*}(0)$ is linear in f, the assembly can be probed one eta monomial at a time — each monomial of the $X_0^{15}(2)$ forms is itself a weakly holomorphic weight-1/2 form of the same multiplier character, holomorphic at every intermediate cusp, and there are 86 of them, extended to 158 by monomials appearing in no form of the panel (`cusp5.m`–`cusp7.m`). On this space the functional does *not* factor through the ∞-side principal part (residual 2.08): the 0-side principal part genuinely enters, and the joint $(\infty, 0)$-side factorisation holds to 10^{-11} with eighty independent quotient conditions. The ledger's "0-block weight 0" finding, and the ∞-only shape of (5) itself, are therefore *representatives* of the true functional modulo the relation ideal of the form space: the joint principal-part coordinates of this character's forms span only 78 of the 128 channel slots, and the 50 leftover combinations are identities of the space, unremovable by further probes. This is the structural origin of the $w_\square$ aliasing reported above, and it sharpens what the derivation must produce: the functional as an element of the quotient, with the square-class channel as its panel-span shadow.

9. The Eisenstein part as ternary theta arithmetic: the s-law and its Siegel–Weil origin

The assembly of the previous subsection makes the functional a finite exact pairing against the cusp constants of f; the derivation it calls for has since been carried out in measured stages, each preregistered before the next base was computed. We record here the state of that campaign. Throughout, $D = qs$ with $q > s$ its two prime factors, and $\mathrm{Gr}(D', R)$ denotes the *genus* of the ternary lattice $\{\alpha \in \mathcal{O}' : \mathrm{tr}\,\alpha = 0\}$ with the reduced-norm form, $\mathcal{O}'$ an Eichler order of level R in the *definite* quaternion algebra of discriminant D' (for $D'R = DN$ this is the Jacquet–Langlands partner of (D, N)); $\bar{\theta}(D', R)$ is its mass-weighted average theta series and $\Theta(D', R) = \mathrm{mass} \cdot \bar{\theta}$ the corresponding genus sum.

Remark 9.1 (the tracked coset: what is arithmetic and what is normalisation)**.** Every fit below reads the ρ-vector of Theorem 8.3 at *one* isotropic coset, and that choice is a normalisation which must be named. At $N > 1$ the coset is a nonzero η^* — the functional of Observation 5.3, well defined because all $2N-2$ choices agree (Observation 5.2); at $N = 1$ there is no such coset, since $2N-1 = 1$ leaves only $\eta = 0$, so the $N = 1$ bases fit a different linear functional. This is the *only* difference between the two regimes. Forcing the zero coset at an $N > 1$ base reproduces the entire $N = 1$ shape — the third term drops, the remaining pair doubles, and the weight sum moves from 0 to 1:

base	fit of E_{eis} mod cusp	$\sum w$
$X_0^{15}(2),\ \eta^*$ (default)	$-\frac{1}{2}\bar\theta(5,2) + \bar\theta(5,6) - \frac{1}{2}\bar\theta(30,1)$	0
$X_0^{15}(2),\ \eta = 0$	$-\bar\theta(5,2) + 2\,\bar\theta(5,6)$	1
$X_0^{22}(3),\ \eta^*$ (default)	$-\bar\theta(11,3) + \frac{3}{2}\bar\theta(11,6) - \frac{1}{2}\bar\theta(66,1)$	0
$X_0^{22}(3),\ \eta = 0$	$-2\,\bar\theta(11,3) + 3\,\bar\theta(11,6)$	1

At $X_0^{22}(3)$ the zero-coset fit is literally $(-2, +3)$, the weights of every $N = 1$ base recorded below. In particular the third term does not vanish at $N = 1$ for the reason one would guess: at $X_0^{15}(2)$ the slot $(30, 1)$ is definite, of one class, and it still drops under $\eta = 0$. Two consequences, which we apply throughout. Absolute weights are comparable only *within* a regime; the weight formulas of Observation 9.3 are those of the η^* normalisation. And the convention-free content of these fits is the mass *ratio* of Observation 9.5, unchanged by any rescaling — which is why it, alone among the readings, survives across both regimes.

Observation 9.2 (the weight-3/2 residue pairing)**.** On every base measured, the class-assembled ρ-vector of Theorem 8.3 lies in the span of the 0-cusp constant vectors of the holomorphic weight-3/2 eta monomials with the conjugate panel character (residuals $\sim 10^{-42}$ at working precision 80), so the functional is represented by a weight-3/2 form E^*; its Eisenstein component $E_{\text{eis}} \in M_{3/2}(\Gamma_0(4DN))$ is exactly rational. Three constants are shared by all nine bases of Observation 9.3: the Eisenstein dimension of $M_{3/2}(\Gamma_0(4DN))$ is 11, the rank of the constants map is 7, and the aliasing kernel of the Gross-average fit has dimension 6. They are constants of that family of levels rather than of the phenomenon: the aliasing is 25 at $X_0^{210}(1)$ and $X_0^{330}(1)$; the triple degenerates to $(5, 3, 1)$ at the four small bases $D = 2q$, $N = 1$ of Observation 9.7 (levels 40–104); and the odd-level family of Observation 9.8 has its own uniform pair, Eisenstein dimension 7 and constants-rank 7, at all four of its levels $60, 84, 132, 140$.

Observation 9.3 (the s-law; nine bases, all with $N > 1$)**.** On every base, in the η^* normalisation of Remark 9.1, the Eisenstein component E_{eis} of Observation 9.2 lies in the span of the $\bar\theta(D', R)$ over the definite structures with $D'R \mid DN$, modulo cusp forms, with support $\{(q, N),\ (q, sN),\ (DN, 1)\}$ and weights

$$w_1 = -\frac{1}{s-1}, \qquad w_2 = \frac{\psi(s)}{2\varphi(s)} = \frac{s+1}{2(s-1)}, \qquad w_3 = -\frac{1}{2},$$

depending on the base only through s. The nine bases are $X_0^{15}(2)$, $X_0^{21}(2)$, $X_0^{33}(2)$ ($s = 3$: weights $-\frac{1}{2}, 1, -\frac{1}{2}$); $X_0^{22}(3)$, $X_0^{34}(3)$, $X_0^{10}(7)$ ($s = 2$: $-1, \frac{3}{2}, -\frac{1}{2}$); $X_0^{35}(2)$, $X_0^{55}(2)$ ($s = 5$: $-\frac{1}{4}, \frac{3}{4}, -\frac{1}{2}$); and $X_0^{77}(2)$ ($s = 7$: $-\frac{1}{6}, \frac{2}{3}, -\frac{1}{2}$). The law was found on the first six and *preregistered* on the last three, each of which extended its range ($s = 3$ at a new q; $s = 5$ at a new q; $s = 7$ new): all three hit. Two simpler laws were killed by preregistered bases along the way: weights following N alone (killed by $X_0^{35}(2)$), and weights following $r = DN/q$ (killed by $X_0^{10}(7)$, which also fixed q as the largest *ramified* prime: there $N = 7 > q = 5$ and the support remains at $(5, *)$).

Observation 9.4 (Siegel–Weil identification)**.** On $X_0^{15}(2)$ the identification is complete. The 16 ternary lattices whose thetas span the relevant subspace of $M_{3/2}(\Gamma_0(60))$ fall into 13 genera, and each genus in the support is quaternionic: the assignment $(D', R) \mapsto \mathrm{Gr}(D', R)$ sends the four definite factorizations $30 = D'R$ bijectively onto the four genera of discriminant 900, distinguishes same-discriminant genera by which prime ramifies (e.g. $\mathrm{Gr}(2,5)$ and $\mathrm{Gr}(5,2)$ are the two genera of discriminant 100), and is inert under level structure at an already-ramified prime. The independent decomposition of E_{eis} over the 13 genus averages plus cusp forms reproduces the fitted weights term by term: $E_{\mathrm{eis}} = -\frac{1}{2}\,\bar\theta(5,2) + \bar\theta(5,6) - \frac{1}{2}\,\bar\theta(30,1)$ modulo cusp forms. The s-law weights are coefficients of genuine Siegel–Weil genus averages, not of single-class thetas.

Observation 9.5 (the mass identities; the s-dependence is normalisation)**.** On all nine bases the genus masses of the support obey

$$\mathrm{mass}(q,N) = \frac{(q-1)(N+1)}{96}, \qquad \frac{\mathrm{mass}(q,sN)}{\mathrm{mass}(q,N)} = \frac{s+1}{2}, \qquad \mathrm{mass}(DN,1) = \frac{(q-1)(s-1)(N-1)}{192},$$

so that, over the genus *sums*, the s-dependence of the weights disappears:

$$E_{\mathrm{eis}} \;=\; \frac{96}{(q-1)(s-1)(N+1)}\Big[-\Theta(q,N) + \Theta(q,sN) - \frac{N+1}{N-1}\,\Theta(DN,1)\Big] \quad \text{mod cusp forms},$$

an identity verified on all nine bases (in the η^* normalisation); the second factor of the middle mass ratio is multiplicative, $\prod_{p|s}(p+1)/2$, over the primes of s. The bracket depends on the base only through N, and the fitted "s-law" is exactly the mass normalisation of a fixed Siegel–Weil combination.

The scale-free part of this,

$$\Big|\frac{w_2}{w_1}\Big| \;=\; \frac{\mathrm{mass}(q,sN)}{\mathrm{mass}(q,N)},$$

is the one statement here untouched by Remark 9.1, and it is the widest-held: it holds on all *nineteen* bases measured — the nine above, the two composite-D bases, the four $N=1$ bases and the four odd-level bases below — in both normalisations, in both parities of DN, and, where the support is degenerate, on both competing supports simultaneously (Observations 9.7 and 9.8). It is therefore the form in which we expect these fits to enter a Siegel–Weil derivation.

Remark 9.6 (preregistration: the first composite s)**.** The base $X_0^{210}(1)$ ($q=7$, $s=30$, $N=1$) separates three extensions of the law that agree on all nine prime-s bases, and we record their predictions in advance of the measurement, which is in flight at the time of writing. (A) The reciprocal reading of Observation 9.3 verbatim: $w_1 = -\frac{1}{29}$, $w_2 = \frac{31}{58}$, $w_3 = -\frac{1}{2}$. (B) The ψ/φ reading (ψ the Dedekind psi): $w_1 = \frac{1}{2} - \psi(30)/2\varphi(30) = -4$, $w_2 = \psi(30)/16 = \frac{9}{2}$. (C) The mass reading of Observation 9.5: $\mathrm{mass}(7,30)/\mathrm{mass}(7,1) = 9 = \prod_{p|30}\frac{p+1}{2}$, hence $w_1 = -\frac{1}{29}$, $w_2 = +\frac{9}{29}$, and *no third term*: the structure $(210,1)$ is indefinite (four ramified places would be required), and its formal mass $(q-1)(s-1)(N-1)/192$ vanishes at $N=1$. Reading (C) is the only one coherent about the third term; readings (A) and (C) agree on w_1. The discriminating ratio is $|w_2/w_1| = \frac{31}{2},\ \frac{9}{8},\ 9$ respectively.

Scorecard: measured, same day. The verdict refutes all three readings of Remark 9.6 *as stated*, in an instructive way: their shared support assumption was wrong. The measurement (residual $1.0\cdot 10^{-42}$, rank 100 of 100, exact rational recognition) gives

$$E_{\mathrm{eis}} \;=\; -2\,\bar\theta(105,1) + 3\,\bar\theta(105,2) \quad \text{mod cusp forms},$$

and the support is genuine, not a pivot artifact of the 25-dimensional aliasing: E_{eis} lies in the span of $\{\bar\theta(105,1), \bar\theta(105,2)\}$ plus cusp forms, and does *not* lie in the span of $\{\bar\theta(7,1), \bar\theta(7,30)\}$ plus cusp forms (rank 87 against 88; but see Observation 9.9 — $(7,1)$ has one prime, and the three-prime competitors are not excluded). Thus at a four-prime discriminant the law re-splits $D = 210$ as $q = 105$ — the full odd ramified part, itself a definite discriminant — and $s = 2$: the weights are twice the prime-s law at $s = 2$, the mass-ratio law of Observation 9.5 extends to ten bases ($\mathrm{mass}(105,1) = \frac{1}{4}$, $\mathrm{mass}(105,2) = \frac{3}{8}$, ratio $\frac{3}{2} = |w_2/w_1|$), and the third term is absent exactly as reading (C) predicted — though for the wrong reason: (C) attributed the absence to the indefiniteness of $(210,1)$, and Remark 9.1 exhibits a *definite* third slot, $(30,1)$ at $X_0^{15}(2)$, dropping under the same normalisation. The overall scale (-2 against the prime-case $-1/(s-1) = -1$) looked at this point like a second law to be found; Remark 9.1 shows it is not arithmetic at all — $X_0^{210}(1)$ has $N = 1$, hence is normalised at $\eta = 0$, and the same doubling appears at $N > 1$ the moment that coset is forced. What does remain a constraint on the derivation is the composite-D' genus mass, which carries a factor $2^{\omega(D')-1}$ relative to the multiplicative prime-case formula of Observation 9.5.

Observation 9.7 (the eleventh base, and the four $N = 1$ bases)**.** Five further measurements, in two preregistered groups, complete the picture. (i) $X_0^{330}(1)$ ($q = 11$, $s = 30$; $M = 660$, residual $1.7 \cdot 10^{-42}$) hits all four predictions recorded in advance — support $\{(165,1), (165,2)\}$ with $q = D/2$ the odd ramified part and no third term, weights $(-2, +3)$, mass ratio $\frac{3}{2}$ from the advance-computed $\mathrm{mass}(165,1) = \frac{5}{12}$, $\mathrm{mass}(165,2) = \frac{5}{8}$ (the same $2^{\omega-1}$ factor), and rank 15 with aliasing 25. Its falsifier fired: the competing largest-prime support $\{(11,1), (11,30)\}$ is excluded (rank 135 against 136), while the winner carries E_{eis} with aliasing dimension 0 — the weights are pinned with no gauge freedom at all. (That competitor has a *single* prime; Observation 9.9 shows the three-prime competitors, never tested at the time, are *not* excluded, so this exclusion does not establish a choice of q.) We record that 330 cannot separate "odd part" from "largest definite divisor": those coincide on every even D, so $X_0^{330}(1)$ replicates the $X_0^{210}(1)$ re-split at fresh primes rather than testing it. (ii) Composite q and $N = 1$ are perfectly confounded across all eleven bases above; the missing cell is prime q with $N = 1$, i.e. $D = 2q$ at $M = 20, 28, 44, 52$. At $q = 5, 7, 11, 13$ (residuals $\sim 10^{-44}$) one finds $E_{\mathrm{eis}} = -2\,\bar\theta(q,1) + 3\,\bar\theta(q,2)$ modulo cusp forms, identical to the composite bases: the doubling does not track composite q. Here the support is genuinely degenerate — the competitor $\{(2,1), (2,q)\}$ also carries E_{eis}, with weights $\left(-\frac{2}{q-1}, \frac{q+1}{q-1}\right)$ — and the mass-ratio law holds on *both* supports at all four bases ($\frac{3}{2}$ on the canonical one; $3, 4, 6, 7$ on the competitor, matching its masses), so it does not depend on the choice of q as the largest ramified prime.

These six $N = 1$ bases are what forced Remark 9.1: the doubling they exhibit survived two successive explanations (composite D, then $N = 1$ itself) before the direct test — forcing $\eta = 0$ at $X_0^{15}(2)$ and $X_0^{22}(3)$ — identified it as the tracked coset and dissolved it. What looked at this stage like the surviving arithmetic — the support, including the re-split $q = D/2$ at four-prime D, together with the mass-ratio law — is reduced further below: Theorem 9.10 shows the support is not determined at all, and that the mass-ratio law is one half of a single identity.

One confound remained. All six of those bases have $s = 2$ — the four small ones outright, $X_0^{210}(1)$ and $X_0^{330}(1)$ after the re-split — so "the $N = 1$ weights are the constant $(-2, +3)$" and "the $N = 1$ weights are twice the s-law" agree on every one of them and nowhere else. Separating them needs $N = 1$ with $s > 2$, hence odd D, hence odd DN: a regime with level $4DN$ rather than $2DN$, which the computation had never entered. It turns out to require only that the 2-part of the discriminant form, there a single $\mathbb{Z}/2$ of odd type rather than $(\mathbb{Z}/2)^3$, be admitted into the closed form of Theorem 8.3; the four smallest such bases are then cheap.

Observation 9.8 (the odd-level quartet; the $N = 1$ weights follow s)**.** At $X_0^{15}(1)$, $X_0^{21}(1)$, $X_0^{33}(1)$ ($s = 3$) and $X_0^{35}(1)$ ($s = 5$), all preregistered, the residue pairing holds (residuals $8.0 \cdot 10^{-42}$, $6.1 \cdot 10^{-41}$, $3.2 \cdot 10^{-42}$, $9.7 \cdot 10^{-43}$) and

base	E_{eis} mod cusp	$\lvert w_2/w_1\rvert$
$X_0^{15}(1)$	$-\bar\theta(5,1) + 2\,\bar\theta(5,3)$	2
$X_0^{21}(1)$	$-\bar\theta(7,1) + 2\,\bar\theta(7,3)$	2
$X_0^{33}(1)$	$-\bar\theta(11,1) + 2\,\bar\theta(11,3)$	2
$X_0^{35}(1)$	$-\frac{1}{2}\bar\theta(7,1) + \frac{3}{2}\bar\theta(7,5)$	3

each restricted support carrying E_{eis} with aliasing dimension 0. These are $2\big(-\frac{1}{s-1}, \frac{s+1}{2(s-1)}\big)$ exactly, so the $N = 1$ weights follow s and the constant reading is refuted. Two further facts, neither predicted. First, the support is *degenerate*: the other ramified prime also carries E_{eis}, at aliasing dimension 0, with weights obeying the same formula read with its own (q', s') — $(3,5) \mapsto (-\frac{1}{2}, \frac{3}{2})$, $(3,7) \mapsto (-\frac{1}{3}, \frac{4}{3})$, $(3,11) \mapsto (-\frac{1}{5}, \frac{6}{5})$, $(5,7) \mapsto (-\frac{1}{3}, \frac{4}{3})$. The law is thus not attached to a choice of q at all when D has two primes. Second, the mass-ratio identity holds on all eight of these fits, canonical and competing, from masses computed in advance; it now stands at *nineteen* bases, across both normalisations, both parities of DN, and every support that carries E_{eis}.

The degeneracy at two primes appeared to locate where the re-split rule has content: a two-prime discriminant cannot select a support, but at $X_0^{210}(1)$ and $X_0^{330}(1)$ a competing span genuinely fails the rank test, so four primes looked as though they did select. That reading was wrong, and the error is instructive: *one* competitor was tested per base, and in both cases it was a D' with a *single* prime ($\{(7,1),(7,30)\}$ at 210, $\{(11,1),(11,30)\}$ at 330). Those exclusions are correct and reproduce here; they are simply not the discriminating test they were taken for. The composite competitors were never run.

Observation 9.9 (the support is never selected)**.** Testing *every* candidate support $\{(D',1),(D',DN/D')\}$ over the definite D' (an odd number of primes) gives the same verdict at all three four-prime bases:

base	D' with one prime	D' with three primes
$X_0^{210}(1)$	$2,3,5,7$ fail	$30,42,70,105$ all carry E_{eis}
$X_0^{330}(1)$	$2,3,5,11$ fail	$30,66,110,165$ all carry E_{eis}
$X_0^{1155}(1)$	11 fails	$105,165,231,385$ all carry E_{eis}

each with the weights $\big(\frac{1}{1-r}, \frac{r}{r-1}\big)$ read off its *own* mass ratio r, and each at aliasing dimension 0. What the rank test separates is one prime from three, not one composite from another. The confirmations are by the same q-expansion machinery used throughout: at $X_0^{330}(1)$ the three supports never previously tested return $-\frac{1}{5}\Theta(30,1) + \frac{6}{5}\Theta(30,11)$, $-\frac{1}{2}\Theta(66,1) + \frac{3}{2}\Theta(66,5)$ and $-\Theta(110,1) + 2\,\Theta(110,3)$, all at rank 135 against 135.

So "$q =$ the larger ramified prime", "odd part of D", "drop the smallest prime" and "largest definite divisor" are not competing rules that $D = 1155$ was to adjudicate. They are four names for one object, and the reason is arithmetic rather than accidental.

Theorem 9.10 (the Eisenstein part, and why no re-split rule exists)**.** *Let D be squarefree with ω prime factors and $N=1$. For any D' a product of $\omega-1$ of them, with $s=D/D'$,*

$$\rho \;=\; \frac{\Theta(D',s)-\Theta(D',1)}{\mathrm{mass}(D',s)-\mathrm{mass}(D',1)},$$

and neither side depends on which prime is s.

Proof. Both sides factor over primes: the Weil representation is multiplicative on orthogonal sums and $(\rho_A\otimes\rho_B)_{00}=(\rho_A)_{00}(\rho_B)_{00}$, and at $N=1$ the tracked coset is $\eta=0$, so ρ is exactly such an entry (Remark 9.1).

At each $p\mid DN$ the local discriminant group is $(\mathbb{Z}/p)^2$, and the two relevant forms are the *hyperbolic* plane $\bar{Q}(b,c)=-bc/p$ (Eichler level p, from $Q=-x^2-pyz$) and the *anisotropic* plane $\bar{Q}(b,c)=(uc^2-b^2)/p$ with u a nonsquare unit (ramified, from $\mathrm{nrd}=-ux^2-py^2+upz^2$). Their Gauss sums are p and $(u\mid p)(-1\mid p)g(1)^2=-p$, so the signatures are 0 and 4 and, with c the lower-left entry of the coset representative,

$$c_p^{\mathrm{Eich}}=\begin{cases}1 & p\mid c\\ +1/p & \text{else,}\end{cases}\qquad c_p^{\mathrm{ram}}=\begin{cases}1 & p\mid c\\ -1/p & \text{else,}\end{cases}$$

whence $c_p^{\mathrm{ram}}=\varepsilon_p\,c_p^{\mathrm{Eich}}$ with $\varepsilon_p=\pm1$. Setting $f_p=(p-1)c_p^{\mathrm{ram}}$ and $g_p=\frac{p+1}{2}c_p^{\mathrm{Eich}}-1$, both cases give $g_p=\frac{p-1}{2}\varepsilon_p c_p^{\mathrm{Eich}}$ and $f_p=(p-1)\varepsilon_p c_p^{\mathrm{Eich}}$, so

$$g_p=\tfrac{1}{2}f_p\qquad\text{at every prime, including } p=2.$$

The two lattices of a support differ at exactly one prime, so with $C=48\cdot 2^{\omega-2}$,

$$\mathrm{mass}_s\,\mathrm{ct}_s-\mathrm{mass}_1\,\mathrm{ct}_1=\frac{c_2}{C}\prod_{p\neq s}f_p\cdot g_s=\frac{c_2}{2C}\prod_{p\mid D}(p-1)\prod_{p\mid D}c_p^{\mathrm{ram}},$$

while $\mathrm{mass}(D',s)-\mathrm{mass}(D',1)=\mathrm{mass}_1(s-1)/2=\prod_{p\mid D'}(p-1)(s-1)/(2C)$, and this *telescopes*: D' and s partition the primes of D, so $\prod_{p\mid D'}(p-1)(s-1)=\prod_{p\mid D}(p-1)$, independent of which prime is omitted. Dividing, the right-hand side is $c_2\prod_{p\mid D}c_p^{\mathrm{ram}}$.

For the left-hand side, B_D is ramified exactly at the primes of D, and ramification is a *local* condition — it does not see the archimedean place — so the (indefinite) Shimura curve lattice carries the same anisotropic plane at each $p\mid D$ as the definite Gross lattices do, and $\rho=c_2^L\prod_{p\mid D}c_p^{\mathrm{ram}}$. The 2-parts agree ($\mathbb{Z}/2$ with $4\bar{Q}=1$ for odd D; the same $(\mathbb{Z}/2)^3$ otherwise), so $c_2^L=c_2$ and the two sides coincide. ☐

The one-prime failures follow from the same computation: there the partner is Eichler at $\omega-1$ primes at once, so the bracket is $\prod_q\big(\frac{q+1}{2}c_q^{\mathrm{Eich}}\big)-1$, which does not factor into g_q's. Nothing telescopes and the mass difference takes ω distinct values instead of one.

Theorem 9.10 was then used *predictively*, on bases the computation had never seen and with nothing fitted — no eta pool, no constant-term table, no least squares, both sides evaluated from the lattices alone. At $D=39,51,55$ (odd) and $D=46,390$ (even) all ten support-instances agree with ρ to $\sim 10^{-120}$; at $D=390=2\cdot3\cdot5\cdot13$ the four supports have mass pairs $(\frac{1}{2},\frac{3}{4})$, $(\frac{1}{4},\frac{1}{2})$, $(\frac{1}{8},\frac{3}{8})$, $(\frac{1}{24},\frac{7}{24})$ and the identical difference $\frac{1}{4}$ — the telescoping in the output rather than on paper.

Finally, the theorem explains the "no third term" of Remark 9.1. A third slot $\Theta(DN,1)$ exists precisely when DN is itself a definite discriminant, i.e. when $\omega(DN)$ is odd; and it is present in exactly those cases, always with coefficient $-\frac{1}{2}$. Across the twelve bases measured, $\omega(DN)=3$ at $X_0^{15}(2)$, $X_0^{21}(2)$, $X_0^{22}(3)$, $X_0^{33}(2)$, $X_0^{35}(2)$, $X_0^{55}(2)$, $X_0^{77}(2)$, $X_0^{10}(7)$, $X_0^{34}(3)$ — all carry the third term — and $\omega(DN)\in\{2,4\}$ at $X_0^{15}(1)$, $X_0^{210}(1)$, $X_0^{330}(1)$, none of which do.

At $N=1$ the algebra B_D is indefinite, which forces $\omega(D)$ even; the absence of a third term at $N=1$ is therefore a corollary of indefiniteness, not an independent fact.

10. The correction determined, at every level

Theorem 9.10 is the $N=1$ case of a single identity that also covers the $N>1$ fits of Observations 9.3 and 9.5, at any tracked coset. Throughout, $\operatorname{mass}(\Delta,R)=\operatorname{mass}(\Delta,1)\prod_{p\mid R}(p+1)/2$ is multiplicative in the level, as in Observation 9.5.

Theorem 10.1 (the Eisenstein part at every level)**.** *Let D be squarefree with $\omega(D)$ even, write $D=D's$ with s any one of its primes, and let N be squarefree with $\gcd(D,N)=1$. Let μ be an isotropic coset of the discriminant form of the Shimura curve lattice of (D,N), and put $S=\{p\mid N:\mu_p\neq 0\}$. With*

$$\operatorname{Tel}_s(\Delta;R)\ =\ \frac{\Theta(\Delta,Rs)-\Theta(\Delta,R)}{\operatorname{mass}(\Delta,Rs)-\operatorname{mass}(\Delta,R)},$$

the ρ-vector read at μ satisfies, at every coset,

$$\rho_\mu\ =\ 2^{-|S|}\sum_{T\subseteq S}(-1)^{|T|}\operatorname{Term}(T),\qquad \operatorname{Term}(T)=\begin{cases}\bar\theta\big(D's\prod T,\ N/\prod T\big) & |T|\ \textit{odd},\\ \operatorname{Tel}_s\big(D'\prod T;\ N/\prod T\big) & |T|\ \textit{even},\end{cases}$$

and the right-hand side does not depend on which prime is s. The case $S=\varnothing$ is Theorem 9.10.

Proof. Beyond the ingredients of Theorem 9.10 the proof needs two things.

The prime factorisation is exact, and its global factor does not see the lattice. The closed form of Theorem 8.3 splits as $\operatorname{ct}_L=\operatorname{glob}(\gamma)\prod_p\operatorname{loc}_p^L(\gamma)$, where $\operatorname{glob}=u\cdot(-a\mid|c|)\cdot(-a,c)_\infty$ collects the Kubota sign and the Kronecker and Hilbert symbols, and $\operatorname{loc}_p=e(-\sigma_p/4)\,G_p\sqrt{|A_p[c]|/|A_p|}$ collects the local signature, Gauss sum and index. Only loc_p depends on L, so telescopings between lattices may be carried out prime by prime.

The tracked coset contributes an indicator, not a phase. The coset enters the closed form only through the solvability of $c\,y=\mu$ and the resulting phase $e(ac\,Q(y))$. The hyperbolic plane at p has exponent p, so a solution exists iff no $p\in S$ divides c; and then $y=c^{-1}\mu$ has $Q(y)=c^{-2}Q(\mu)=0$ because μ is *isotropic*. Hence $\rho_\mu=\kappa\,\rho_0$ with $\kappa=\prod_{p\in S}[p\nmid c]$, with no phase at all.

Now $\rho_0=\prod_{p\mid D}c_p^{\mathrm{ram}}\prod_{p\mid N}c_p^{\mathrm{Eich}}$, and the two local values displayed in the proof of Theorem 9.10 give, at every prime, $\frac12\big(c_p^{\mathrm{Eich}}-c_p^{\mathrm{ram}}\big)=c_p^{\mathrm{Eich}}\big[p\nmid c\big]$. Therefore

$$\rho_\mu=\prod_{p\mid D}c_p^{\mathrm{ram}}\prod_{p\mid N,\,p\notin S}c_p^{\mathrm{Eich}}\prod_{p\in S}\tfrac12\big(c_p^{\mathrm{Eich}}-c_p^{\mathrm{ram}}\big),$$

and expanding the last product over subsets $T\subseteq S$ gives $2^{-|S|}(-1)^{|T|}$ times $\prod_{p\mid D}c_p^{\mathrm{ram}}\prod_{p\in T}c_p^{\mathrm{ram}}\prod_{p\mid N,\,p\notin T}c_p^{\mathrm{Eich}}$. It remains to identify that with $\operatorname{Term}(T)$. If $|T|$ is odd then $\omega(D's\prod T)=\omega(D)+|T|$ is odd, the algebra of that discriminant is definite, and $\operatorname{Gr}(D's\prod T,\ N/\prod T)$ — ramified at the primes of D and of T, Eichler at the remaining primes of N — has exactly that constant term. If $|T|$ is even that discriminant is indefinite and no such genus exists; instead the two levels $N/\prod T$ and $(N/\prod T)s$ over $D'\prod T$ differ at s alone, all common factors cancel from the quotient because the mass is multiplicative in the level, and $g_s=\frac12 f_s$ restores the missing c_s^{ram} exactly as in Theorem 9.10 — which is also why the answer is independent of s. Finally, as at $N=1$, the 2-adic normalisations of the lattices involved must agree after telescoping; the per-prime factorisation exhibits this directly (at $X_0^{22}(3)$ and c prime to the level, $\operatorname{loc}_2=(1+i)/4$ for the

Shimura lattice and for $\mathrm{Gr}(66,1)$ alike, and the telescoped 2-part of the $\mathrm{Gr}(11,\cdot)$ pair is the same). $\square$

Three consequences are worth recording. First, Theorem 10.1 promotes Observation 9.5 to an identity: for N prime and $\mu \neq 0$ the set $S = \{N\}$ has two subsets, and the theorem reads $\rho_\mu = \frac{1}{2}\mathrm{Tel}_s(D'; N) - \frac{1}{2}\bar{\theta}(DN, 1)$, which is the displayed formula of that observation term for term, including the coefficient $-\frac{N+1}{N-1}$ of the third slot. Second, the right-hand side is symmetric in $D' \leftrightarrow s$, so at $N > 1$ as at $N = 1$ *the support is not selected*: both choices of which prime of D plays the role of s give the same Eisenstein series. This is Observation 9.9 again, now at every level and with a reason.

Third, the theorem settles the coset bookkeeping that Observation 5.2 could only record. For μ isotropic with support $S = \{p \mid N : \mu_p \neq 0\}$ it gives

$$\rho^S \;=\; \rho^0 \cdot \prod_{p \in S} [\, p \nmid c \,],$$

since $\prod_{p\in S} \frac{1}{2}(c_p^{\mathrm{Eich}} - c_p^{\mathrm{ram}}) = \prod_{p \in S} c_p^{\mathrm{Eich}}[p \nmid c]$ and the remaining factors reassemble $\mathrm{ct}_L(\gamma, 0)$. So distinct tracked cosets do not give unrelated functionals: they give one vector restricted to nested sets of cosets, and the functional depends on μ only through S. Consequently the number of isotropic cosets is $\prod_{p|N}(2p-1)$, those with support exactly S number $\prod_{p\in S}(2p-2)$, and the number of *distinct* functionals is $2^{\omega(N)} - 1$. For N prime that last count is 1 — which is exactly the agreement Observation 5.2 records, now with a reason — while for composite N it is not, and the normalisation $4(N-1) = 2(2N-2)$ implicit in the equivalent form $\mathrm{mult} = \sum_\eta c_\eta(0)/(4(N-1))$ is correct only when $\omega(N) = 1$; in general

$$\sum_{\eta \neq 0} \rho^{\,\eta} \;=\; \rho^0 \Big(\prod_{p|N} \big(1 + (2p-2)[p \nmid c]\big) - 1 \Big).$$

Measured at $X_0^{22}(3)$ ($\omega = 1$: one functional, all four nonzero cosets agreeing), $X_0^{35}(6)$ and $X_0^{15}(14)$ ($\omega = 2$: three functionals each, class sizes $2, 4, 8$ and $2, 12, 24$), with the restriction law holding at 4, 14 and 38 of 4, 14 and 38 cosets. Which of the $2^{\omega(N)} - 1$ functionals would be the multiplier at composite N is not decided here, and cannot be until a composite-N base with independent ground truth exists.

The identity was checked at every coset of the level $4DN$, with nothing fitted — both sides evaluated from the lattices alone. On the eight banked $N > 1$ bases the derived weights agree with the fitted ones exactly (worst residual $3 \cdot 10^{-119}$). It was then used *predictively* at eight bases that had never been looked at — $X_0^6(5)$, $X_0^{10}(3)$, $X_0^{14}(3)$, $X_0^6(7)$, $X_0^{14}(5)$, $X_0^{26}(3)$, $X_0^{39}(2)$, $X_0^{34}(3)$ — with *both* choices of s at each: sixteen instances, all confirmed to $\sim 10^{-119}$, with weights ranging from $-\frac{1}{16}$ to $\frac{3}{2}$; dropping the third term fails at relative size 1.4 to 3.5 at every one of them. Composite N, where $|S|$ can be 2 and the combination has four terms, was checked at $X_0^{35}(6)$ and $X_0^{15}(14)$ at *every* nonzero isotropic coset rather than the default one — 14 and 38 of them, both choices of s — for a further 112 confirmed instances. Finally $\omega(D) = 4$ with $N > 1$, whose smallest instance is $X_0^{210}(11)$ at level 9240 with 27648 cosets and a discriminant group of order $2.9 \cdot 10^7$: both $s = 2$ and $s = 3$, two tracked cosets each, all confirmed to $2 \cdot 10^{-118}$.

For the combination of Theorem 10.1 to be usable as an E in what follows, it must be a holomorphic form of weight $\frac{3}{2}$ on $\Gamma_0(4DN)$ carrying a single multiplier. Both are consequences of the local structure already in hand.

Lemma 10.2 (level and multiplier of the Gross thetas)**.** *Let Δ be squarefree with $\omega(\Delta)$ odd, let R be squarefree and coprime to Δ with $\Delta R \mid DN$, and let L be the trace-zero part of an*

Eichler order of level R in the definite quaternion algebra of discriminant Δ, with $Q = \mathrm{nrd}$. Then

(i) *the Gram matrix has determinant $2(\Delta R)^2$, and the level of Q is $4 \cdot \mathrm{odd}(\Delta R)$, which divides $4DN$;*
(ii) *θ_L is a holomorphic modular form of weight $\frac{3}{2}$ on $\Gamma_0(4DN)$;*
(iii) *all the θ_L occurring in Theorem 10.1 carry the same multiplier, namely the one governing ρ; hence their combination is a modular form, and fE has trivial multiplier for E that combination.*

Proof. (i) By the local analysis in the proof of Theorem 9.10, the discriminant form of L is an anisotropic plane $(\mathbb{Z}/p)^2$ at each $p \mid \Delta$, a hyperbolic plane $(\mathbb{Z}/p)^2$ at each $p \mid R$, and $\mathbb{Z}/2$ with $4\bar{Q} = 1$ at 2 when ΔR is odd (the same $\mathbb{Z}/2$ that makes $c_2^L = c_2^G$ there). Multiplying orders gives $|A_L| = 2(\Delta R)^2 = \det$. The level of an even lattice is the exponent of $\bar{Q}$ as a $\mathbb{Q}/\mathbb{Z}$-valued form: each odd $p \mid \Delta R$ contributes p, and the 2-part contributes 4, since $4\bar{Q} = 1$ there. Hence the level is $4 \cdot \mathrm{odd}(\Delta R)$. As DN is squarefree and $\Delta R \mid DN$, we get $\mathrm{odd}(\Delta R) \mid DN$ and so the level divides $4DN$.

(ii) A positive definite ternary theta series is holomorphic of weight $\frac{3}{2}$ on Γ_0 of its level, and by (i) that group contains $\Gamma_0(4DN)$; the genus average is a convex combination of such.

(iii) The identity of Theorem 10.1 is an identity of Weil-representation entries in closed form, and its proof nowhere uses that γ belongs to any chosen set of coset representatives: it holds for every γ. Now fix $\gamma_0 \in \Gamma_0(4DN)$ and a representative w. Each side is a weight-$\frac{3}{2}$ object slashed by a matrix, so evaluating at $\gamma_0 w$ multiplies the left side by $\nu_{\mathrm{Shim}}(\gamma_0)$ and the right by $\nu_{\mathrm{Gross}}(\gamma_0)$. Since the identity holds at $\gamma_0 w$ as well as at w, and the common value is not identically zero, $\nu_{\mathrm{Gross}} = \nu_{\mathrm{Shim}}$. The same argument applied to the individual terms shows they share that multiplier. Finally the panel character is by definition the one for which E^* reproduces ρ, i.e. the conjugate of ν_{Shim}, so fE has trivial multiplier. $\square$

Each clause was checked. The determinant and level of (i) were computed for all 100 Gross lattices arising at the bases treated here, agreeing with $2(\Delta R)^2$ and $4 \cdot \mathrm{odd}(\Delta R)$ in every case and dividing $4DN$ in every case. For (iii), the identity was re-evaluated *off* the representative list, at $\gamma_0 w$ for four or five $\gamma_0 \in \Gamma_0(4DN)$ against a spread of twelve representatives, at $X_0^{21}(2)$, $X_0^{22}(3)$, $X_0^{55}(2)$, $X_0^{77}(2)$, $X_0^{34}(3)$ and $X_0^{15}(2)$: it holds at all 48–60 matrices per base, to $4 \cdot 10^{-96}$. Independently, at seven of the bases the combination is exhibited inside the space of holomorphic weight-3/2 eta quotients with the panel character, which gives (iii) there directly; at the remaining bases that space is too small to decide, which is why the argument above is the one used.

The multiplicity as a residue pairing. The assembly of §8 expresses the $m = 0$ multiplier as a sum of constant terms of f against the ρ-entries. That sum is one half of a residue identity, and the other half is a pairing against the Fourier coefficients of *any* weight-3/2 form carrying those constants — in particular against the explicit combination of ternary genus thetas supplied by Theorem 10.1. Write $a_n(h \mid w)$ for the coefficient of q^n in the expansion at ∞ of h slashed by a coset representative w.

Theorem 10.3 (the residue pairing)**.** *Let $M \geq 1$, let f be a weakly holomorphic modular form of weight $\frac{1}{2}$ on $\Gamma_0(M)$ with multiplier χ, holomorphic on $\mathbb{H}$, and let E be a* holomorphic *modular form of weight $\frac{3}{2}$ on $\Gamma_0(M)$ whose multiplier is* conjugate *to χ — by which we mean precisely that the product fE has trivial multiplier on $\Gamma_0(M)$. (In half-integral weight this is a condition on the theta factors as well as on χ, and it is verified, not assumed, in the cases* *used below.) Then*

$$\sum_{w\in\Gamma_0(M)\backslash \mathrm{SL}_2(\mathbb{Z})}\ \sum_{n\geq 0} a_{-n}(f \mid w)\, a_n(E \mid w) \ =\ 0,$$

each inner sum being finite. Separating the term $n = 0$,

$$\sum_{w} a_0(f \mid w)\, a_0(E \mid w) \ =\ -\sum_{w}\sum_{n>0} a_{-n}(f \mid w)\, a_n(E \mid w). \tag{$*$}$$

If $a_0(E \mid w) = \rho_w$ for every w, the left-hand side of $(*)$ is the assembly, i.e. $c_{\eta^*}(0)$; and the right-hand side is unchanged if E is replaced by any other holomorphic weight-$\frac{3}{2}$ form with the same constants.

Proof. Put $h = fE$. It is holomorphic on $\mathbb{H}$, of weight 2, and by hypothesis has trivial multiplier, so h is a meromorphic modular form of weight 2 on $\Gamma_0(M)$ with poles only at the cusps; hence $\omega = h(\tau)\, d\tau$ is a meromorphic differential on $X_0(M)$, holomorphic away from the cusps. By the residue theorem on a compact Riemann surface, $\sum_s \mathrm{Res}_s\, \omega = 0$.

Fix a cusp s of width h_s with scaling matrix σ_s, and set $q_s = e^{2\pi i\tau/h_s}$. Then $(h \mid \sigma_s)(\tau) = \sum_n a_n(h \mid \sigma_s) q_s^n$ and $\omega = \frac{h_s}{2\pi i}\big(\sum_n a_n(h \mid \sigma_s) q_s^n\big)\frac{dq_s}{q_s}$, so $\mathrm{Res}_s\, \omega = \frac{h_s}{2\pi i}\, a_0(h \mid \sigma_s)$. The cosets lying over s are $\sigma_s T^j$ for $0 \leq j < h_s$, and slashing by T^j multiplies the n-th coefficient by $e^{2\pi i n j/h_s}$, which fixes $n = 0$; hence each of those h_s cosets contributes the same a_0. Summing over all cosets therefore supplies the width factor automatically:

$$\sum_{w} a_0(h \mid w) \ =\ \sum_{s} h_s\, a_0(h \mid \sigma_s) \ =\ 2\pi i \sum_{s} \mathrm{Res}_s\, \omega \ =\ 0.$$

Finally $h \mid w = (f \mid w)(E \mid w)$, and $E \mid w$ has no negative coefficients while $f \mid w$ has only finitely many, so $a_0(h \mid w) = \sum_{n\geq 0} a_{-n}(f \mid w)\, a_n(E \mid w)$ is a finite sum. This is the displayed identity, and $(*)$ is the case $n = 0$ separated off.

For the last statement, let E' be another such form and $g = E - E'$. Then g is holomorphic of weight $\frac{3}{2}$ with $a_0(g \mid w) = 0$ for every w — a cusp form — so applying the identity to the pair (f, g) makes its $n = 0$ block vanish, leaving $\sum_w \sum_{n>0} a_{-n}(f \mid w) a_n(g \mid w) = 0$. ☐

The independence in the last paragraph is worth isolating, because it is the reason the cuspidal ambiguity of Theorem 10.1 costs nothing. Every method used here determines a weight-3/2 form only through its constant terms, hence only modulo $S_{3/2}$; by Theorem 10.3 that ambiguity is annihilated by the pairing. (This is Borcherds' obstruction principle in the present setting, and it also accounts for the two-sided shape of the pairing: the sum runs over *all* cosets, so principal parts at 0 enter on the same footing as those at ∞.)

Corollary 10.4. *Let E be the combination of genus theta series furnished by Theorem 10.1. Then the $m = 0$ multiplier is*

$$c_{\eta^*}(0) \ =\ -\sum_{w}\sum_{n>0} a_{-n}(f \mid w)\, a_n(E \mid w),$$

a pairing of the principal parts of f against Fourier coefficients of an explicit mass-weighted combination of ternary genus theta series — that is, against representation numbers of definite ternary quadratic forms.

The hypotheses were verified rather than assumed. That fE has trivial multiplier on $\Gamma_0(60)$ is checked directly at $X_0^{15}(2)$, with $f \cdot f$ as a control exhibiting Newman's $(-1/d)$. The vanishing $\sum_w a_0(fE \mid w) = 0$ is confirmed to some 70 digits on all nine panel forms at each of $X_0^{15}(2)$, $X_0^{21}(2)$ and $X_0^{22}(3)$ — the three distinct w_{sq} regimes. And at $X_0^{15}(2)$ the substitution of

Corollary 10.4 was carried out: the genus combination lies in the space of holomorphic weight-3/2 eta quotients of level 60 with the panel character, reproduces ρ at all 288 cosets to $6 \cdot 10^{-77}$ — computed along a different route than Theorem 10.1 uses — and returns the same pairing as the fitted representative on every panel form, to every digit printed, although the two forms differ.

The coefficients in closed form. Corollary 10.4 makes $-a_E(m)$ a combination of representation numbers of definite ternary forms. Siegel–Weil turns each of those into a product of local densities — the terms really are genus averages rather than single-class thetas, which is the content of Observation 9.4 — and the combination collapses, at odd primes, into one elementary factor per prime of DN against a Hurwitz class number.

Proposition 10.5 (the multiplicity coefficients)**.** *Let D be squarefree with $\omega(D) = 2$, let N be 1 or a prime coprime to D (the prime 2 is allowed in either slot), and let $m \geq 1$. Write $-4m = D_0c^2$ with D_0 fundamental, and for a prime p set $k_p = v_p(c)$, $\chi_p = \left(\frac{D_0}{p}\right)$ and $\sigma_j(p) = 1 + p + \cdots + p^j$ (with $\sigma_{-1} = 0$). Then, with H the Hurwitz class number,*

$$-a_E(m) \;=\; 12\, H(4m) \prod_{p|D} \frac{1-\chi_p}{(p-1)\big(\sigma_{k_p} - \chi_p\sigma_{k_p-1}\big)} \prod_{p|N} \frac{(p-\chi_p)\, p^{k_p}}{(p^2-1)\big(\sigma_{k_p} - \chi_p\sigma_{k_p-1}\big)},$$

the second product being empty when $N = 1$. In particular $a_E(m) = 0$ unless every $p \mid D$ is non-split in $\mathbb{Q}(\sqrt{-m})$, i.e. unless $\mathbb{Q}(\sqrt{-m})$ embeds in B_D: the support rule is part of the formula rather than a side condition.

The conductor factors are the only place c enters, and they have the same denominator $\sigma_k - \chi\sigma_{k-1}$ at both kinds of prime, with numerator 1 at a ramified prime and p^k at the Eichler prime. For m squarefree every k_p is 0, both σ-factors are 1, and the statement collapses to

$$-a_E(m) \;=\; H(4m) \prod_{p|D} \frac{1-\left(\frac{-m}{p}\right)}{p-1} \;\cdot\; \frac{12\big(N - \left(\frac{-m}{N}\right)\big)}{N^2-1},$$

which is how it was found.

The squarefree shape was read off at $N = 3$ and $N = 7$ and tested at $N = 5$ (on $X_0^{21}(5)$ and $X_0^{33}(5)$), $N = 11$ and $N = 13$, none of which entered the fit; the conductor factors were then read off at $X_0^{15}(7)$, $X_0^{35}(3)$, $X_0^{21}(5)$ with $k \leq 2$. The full statement holds for *every* $m < 400$ at all six bases — 156 of them non-squarefree — and, pushed to $m < 800$ at $X_0^{15}(7)$ to reach $k = 3$, for all 799 values, 310 of them non-squarefree. So the k-dependence is verified one step beyond where it was fitted.

Two of the regimes deserve a word. The case $N = 1$ is not a limit of the others — the Eichler factor would read $0/0$ — but it is the empty product, and this is where the constant 12 shows itself as global rather than as part of the level factor. It was checked on *all six* bases of the model set with DN odd, namely $D = 15, 35, 39, 55, 65, 87$ (each a product of two primes), with both choices of which prime is s: the residual after dividing out the ramified product and the conductor corrections is exactly 12, for every $m < 250$, in all twelve runs.

Finally $p = 2$, which carries no exception: the two products above are taken over *all* primes of D and of N, the prime 2 included, with the same two expressions. Read off in the three regimes separately — $2 \mid D$ with $N = 1$, $2 \mid D$ with N odd, and $N = 2$ — the 2-adic factors are exactly ram_2 and eich_2: for instance ram_2 gives $\frac{1}{2}, \frac{1}{7}, \frac{1}{11}$ at $(k, \chi) = (1, -1), (2, 0), (3, -1)$ and eich_2 gives $\frac{2}{3}, \frac{4}{9}, \frac{1}{3}$ at $(0, 0), (1, 0), (2, 1)$. The two $2 \mid D$ regimes produce identical tables, as they must if “ramified at 2” is one local factor.

It is worth saying how this looked before the conductor parametrisation, because the appearance of an exception at 2 was an artefact. Restricted to squarefree m and keyed on $m \bmod 8$, the same data reads as $\delta_2 = 1, \frac{1}{2}, 0$ for $m \equiv 1, 2, 5, 6,\ 3,\ 7 \pmod 8$ when $2 \mid D$, and as $8, 8, 6, 8, 8, 4$ when $N = 2$ — neither of which resembles the odd law. Both are just its shadow: for m squarefree, $m \equiv 1, 2 \pmod 4$ forces $k_2 = 0, \chi_2 = 0$, while $m \equiv 3 \pmod 4$ forces $k_2 = 1$ with $\chi_2 = \mp 1$ according as $m \equiv 3, 7 \pmod 8$, and substituting reproduces the tables.

The proposition therefore covers every base in the model set. It was verified on nineteen of them spanning all four regimes — $X_0^{15}(1)$, $X_0^{35}(1)$, $X_0^{39}(1)$, $X_0^{55}(1)$; $X_0^{10}(1)$, $X_0^{14}(1)$, $X_0^{22}(1)$, $X_0^{26}(1)$; $X_0^{15}(2)$, $X_0^{21}(2)$, $X_0^{33}(2)$, $X_0^{35}(2)$; $X_0^{10}(3)$, $X_0^{22}(3)$, $X_0^{10}(7)$, $X_0^{6}(5)$, $X_0^{14}(5)$, $X_0^{34}(3)$, and $X_0^{15}(7)$ — at every $m < 250$: 4731 of 4731 values.

Remark 10.6 (what Table 1 pins down, and what it does not)**.** Proposition 10.5 and Table 1 are two representatives of one functional, and they do *not* agree coefficient by coefficient. At $X_0^{15}(2)$ the nine panel forms touch the six indices $m = 1, 2, 3, 10, 15, 30$, where $\frac{1}{2}\big(-a_E(m)\big)$ reads $0, 0, 1, 2, 1, 2$ against the A_m of Table 1, which read $0, -1, 0, 1, 0, 0$. Both reproduce all nine measured multipliers exactly. They can do so because the panel's matrix of ∞-side principal parts has rank 4 in those six indices, so an ∞-only functional is determined only modulo a 2-dimensional space and the two vectors lie in one coset of it.

Two consequences. First, the rule "the weight vanishes unless $N \mid m$" fixes a representative inside that ambiguity: it is a gauge, not a property of the correction, whose coefficients on that base are nonzero at $m = 3, 7, 13, 15, 27, 33, 37$. What Corollary 3.2 governs is the vanishing of the level-prime Whittaker factor, and — through Proposition 4.1 — which m can contribute a logarithm from $\kappa_\eta(m)$; it does not govern the support of $-a_E$, which is the embedding condition at the primes of D. Second, this is the $w_\square$ aliasing of §8 in the coordinates of the closed form, and it shows that aliasing is not something a derivation could remove: the canonical representative $-a_E$ does not have the two-channel shape of (5) at all, so $w_\square$ is a coordinate of a gauge-fixed representative rather than an invariant. Conjecture 8.1 should be read accordingly.

What the multipliers do pin down is reproduced exactly. Evaluating $\frac{1}{2}\sum_m c_\infty(-m)\big(-a_E(m)\big)$ from Proposition 10.5 alone returns the measured multiplier on all 39 forms carrying one, across the nine bases $X_0^{6}(5)$, $X_0^{6}(7)$, $X_0^{6}(19)$, $X_0^{10}(3)$, $X_0^{10}(11)$, $X_0^{14}(5)$, $X_0^{15}(2)$, $X_0^{22}(7)$ and $X_0^{34}(5)$. This is a longer chain than the sweep above: there the closed form is checked against the theta combination it was derived from, whereas here the right-hand side is built from the closed form alone while the left-hand side is the kernel-consistency or Hauptmodul measurement of §5, which evaluates no Eisenstein series at all. The ∞ block suffices on these panels; on the larger monomial probe space of §8 the 0-side genuinely enters, and there the pairing must be taken over all cosets as in Corollary 10.4.

11. Where the correction comes from

Corollary 4.2 and Observation 5.1 are in direct tension with Observation 5.3 and with the failure of the uncorrected computation on $X_0^{6}(19)$. Two readings suggest themselves, and both can now be excluded.

(1) *The constant term $c_0(0)$ of the form F actually attached to Ψ_F differs from that of the coset sum F_f; the principal parts agree to 10^{-103}, but the principal part alone does not determine $c_0(0)$.* — The premise fails. A weakly holomorphic form of weight $\frac{1}{2}$ for ρ is determined by its principal part up to $M_{1/2}(\rho)$, and $M_{1/2}(\rho) = 0$ for the modules occurring here. This is structural, not a base-by-base observation: invariants correspond to self-dual isotropic subgroups, and the D-part of the discriminant form

is an anisotropic $(\mathbb{Z}/p)^2$ at every $p \mid D$ — which contains no nonzero isotropic vector, so no isotropic subgroup can meet it, no cyclic summand can absorb it, and no $H^\perp/H$ lift can remove it. Hence $F = F_f$ exactly, and $c_0(0)$ is the measured 0.

(2) *The correction compensates an error elsewhere in the computation of the CM values, outside Schofer's formula; being a d-independent linear functional of the principal part, it has the shape of a per-form normalisation.* — A per-form normalisation rescales a whole row of the CM table by one factor, and a uniform rescaling cannot turn a consistent linear system into an inconsistent one. On $X_0^6(19)$ the uncorrected values are mutually inconsistent, and with the correction the published model is recovered. Moreover the correction is now identified rather than fitted: by Theorem 10.3 and Corollary 10.4 it is the residue pairing of the principal parts of f against an explicit holomorphic weight-$\frac{3}{2}$ form, and by Proposition 10.5 its coefficients are a Hurwitz class number times one local factor per prime of DN. That is an arithmetic term of exactly the family Schofer's formula is assembled from, not the signature of a misnormalisation.

What survives is that terms *inside* Schofer's formula were dropped, and his statement locates them precisely. In [9, Thm. 3.3] the coefficient attached to a pair (η, λ) is $\kappa_{\eta,\lambda}(m_1) = \sum_{0 \le m \le m_1} d_{\eta_+ + \lambda_+}(m_1 - m)\, \kappa_{\eta_- + \lambda_-}(m)$, where $d_{\eta_+ + \lambda_+}(m) = \#\{x_1 \in \eta_+ + \lambda_+ + L_+ : Q(x_1) = m\}$ and, at $m = 0$,

$$\kappa_{\eta_- + \lambda_-}(0) \;=\; k_0(0)\, \varphi_{\eta_- + \lambda_- + L_-}(0),$$

a *characteristic function* of the coset. Since L_+ is positive definite, $Q(x_1) = 0$ forces $x_1 = 0$, so $d_{\eta_+ + \lambda_+}(0)$ is 1 when $\eta_+ + \lambda_+ \equiv 0$ and 0 otherwise. Hence

$$\kappa_{\eta,\lambda}(0) \;=\; k_0(0) \cdot \big[\eta_+ + \lambda_+ \equiv 0\big]\big[\eta_- + \lambda_- \equiv 0\big],$$

and the $m = 0$ part of the formula is $k_0(0) \sum_\eta \sum_\lambda c_\eta(0)$ over the pairs satisfying both congruences, with λ running over $L/(L_+ + L_-)$.

The collapse to $c_0(0)k_0(0)$ is therefore valid exactly when $L = L_+ + L_-$. There $\lambda = 0$ is the only class, the two congruences read $\eta_+ \equiv 0$ and $\eta_- \equiv 0$, and only $\eta = 0$ survives. That is the maximal-order situation, and it is the one in which the recipe was checked — Yang's worked examples have $L = L_+ + L_-$. For an Eichler order of level $N > 1$ the index $[L : L_+ + L_-]$ exceeds 1, the λ-sum is nontrivial, and cosets $\eta \neq 0$ whose two components are matched by some λ contribute. Those are the dropped terms; $c_0(0) = 0$ removes the one that was kept, so they carry the whole correction. That the surviving cosets share one constant term, which Observation 5.2 recorded empirically, is now a consequence of N being prime: the number of distinct functionals is $2^{\omega(N)} - 1$, which is 1 exactly when $\omega(N) = 1$.

The scope violation is visible in the sources' own terms. Kudla–Rapoport–Yang state that they consider only Eisenstein series built from sections given by the characteristic functions of the completions of $\mathcal{O}_k$, and note that more general sections would cost "more elaborate calculations of Whittaker functions" and "incorporation of a level structure in the moduli problem"; they also fix a maximal order. At a firing discriminant our L_- is N-scaled, which is such a more general section, so the case-completeness assertion that becomes Schofer's Theorem 4.1 and Yang's Lemma 16 is being applied outside its stated hypotheses.

The vanishing at $\eta \neq 0$ has a second guard in [9], and it fails the same way. Lemma 2.18 there concludes $b_\varphi(0, t) = 0$ for $\varphi \neq \varphi_0$, citing a local statement: $W_{0,p}(s, \varphi_p) = 0$ whenever φ_p is not the characteristic function of the local lattice. For a unimodular L_p that is vacuous — $L_p^\vee/L_p$ is trivial, so the only such φ_p is $\mathrm{char}(L_p)$ and there is nothing else to test. At a p-scaled L_p there is. Evaluating $W_{0,p}$ directly on the cosets of $L_p^\vee/L_p \cong (\mathbb{Z}/p)^2$ for $A = p \cdot (\text{unimodular})$, at a *split* p — the case a firing discriminant forces — gives

$$\#\{\mu \neq 0 : W_{0,p}(0,\mu) \neq 0\} \; = \; 4, \; 8, \; 12 \quad \text{at } p = 3, 5, 7,$$

each with value 1; that is $2(p-1)$, which is exactly the number of nonzero *isotropic* cosets of a hyperbolic plane over $\mathbb{F}_p$. (Lemma 11.1 below sharpens this from a value to an identity of polynomials: the factor is the constant 1, so its s-derivative vanishes too. The distinction matters, and is what fixes where the derivative of the product is taken.) So at the level prime the $m = 0$ local Whittaker is nonzero on precisely the $2N-2$ nonzero isotropic cosets, with equal weight, and Lemma 2.18 sets all of them to zero on the strength of a statement established where no such coset exists. Those are the dropped terms, and their count and equal weighting are exactly the structure the measurements report: all nonzero isotropic η carry the same $c_\eta(0)$, and the multiplier is $\sum_\eta c_\eta(0)/(4(N-1)) = \frac{1}{2}c_\eta(0)$ over the $2N-2$ of them.

This also changes what should replace the dropped term, and the replacement is not $k_0(0)$ times a count. In [9, Lem. 2.18] the $m = 0$ Fourier coefficient is

$$\mathcal{E}_0(\tau, s; \varphi, +1) \; = \; v^{s/2}\varphi(0) \; - \; \sqrt{\pi}\, i\, v^{-s/2}\, \frac{\Gamma\big(\frac{s+1}{2}\big)}{\Gamma\big(\frac{s}{2}+1\big)} \prod_p W_{0,p}(s, \varphi_p),$$

and for a nonzero coset φ_μ one has $\varphi_\mu(0) = 0$, since $0 \notin \mu + L$. The first term therefore drops. The Whittaker product does still vanish at $s = 0$ — though not for the reason one would expect, as the proof of Proposition 11.2 shows — so what survives at $\mu \neq 0$ is its *s-derivative*, rather than the regularised constant $k_0(0)$ that Theorem 3.3 attaches to φ_0. We now evaluate it. The one ingredient not already in hand is the level-prime factor at the *zero* coset, which the recipe of §2 never needs and which the accompanying code cannot return, its defining series failing to terminate. It is fixed by calibrating a normalisation against [9, Lem. 2.18] itself.

Lemma 11.1 (the $m = 0$ local factor of a p-scaled plane)**.** *For a binary $\mathbb{Z}_p$-lattice L and $\varphi = \mathrm{char}(\mu{+}L)$, put $\alpha_k = p^{-k}\#\{x \in (\mu{+}L)/p^kL : Q(x) \equiv 0 \bmod p^k\}$ and $G(X) = \sum_{k\geq 0} \alpha_k X^k$ with $X = p^{-s}$. Then $W_{0,p}(s,\varphi) = (1-X)\,G(X)$, and*

L_p	μ	α_k	$W_{0,p}(s,\varphi)$
unimodular, split	0	$1 + k\frac{p-1}{p}$	$(1-X/p)/(1-X)$
unimodular, inert	0	$1, \frac{1}{p}, 1, \frac{1}{p}, \ldots$	$(1+X/p)/(1+X)$
p-scaled, split	$\neq 0$ *isotropic*	1	1
p-scaled, split	0	$(k-1)(p-1)+p$	$\big(1+(p-2)X\big)/(1-X)$

In particular the last has a simple pole *at $s = 0$: $W_{0,p}(s,\varphi_0) = \frac{p-1}{s\log p} + O(1)$.*

Proof. The two unimodular rows are forced: [9, Lem. 2.18] evaluates $\prod_p W_{0,p}(s,\varphi_0) = c_0\, L(s,\chi_d)/L(s{+}1,\chi_d)$ in the unimodular case, so the local factor there must be the p-factor of that ratio, which is $(1-\chi_d(p)X/p)/(1-\chi_d(p)X)$; the displayed α_k are elementary counts, and $(1-X)G(X)$ returns exactly those two expressions. This determines the normalisation. The third and fourth rows are the same counts for the p-scaled plane $Q = pxy$: at an isotropic $\mu = (a/p, 0)$ with a a unit, $Q(x) \equiv 0$ forces $y \equiv 0$ and $\alpha_k = 1$; at $\mu = 0$ one has $xy \equiv 0 \bmod p^{k-1}$, giving $\alpha_k = (k-1)(p-1)+p$. Summing and multiplying by $1-X$ gives the last two entries. ☐

The third row is a check rather than an input: it is the object the code *can* compute, and it returns the constant 1, in agreement (`km0poly.m`). All four rows were verified against brute-force counts at $p = 3, 5, 7$ (`km0whit.py`).

Proposition 11.2 (the dropped coefficient)**.** *Let N be prime and μ a nonzero isotropic coset. Then*

$$\kappa_\mu(0) \;=\; -\,\frac{\log N}{N-1},$$

and summing over the $2N-2$ such cosets reproduces the correction of Observation 5.3: in the normalisation of §2 the dropped $m=0$ terms contribute $\frac{1}{2}c_\eta(0)\log N$ per CM point, i.e. exactly $\mathrm{mult}(f)\sum_{p\mid N/(N,d)}\log p$.

Proof. At a firing discriminant a nonzero isotropic μ is supported at the level prime alone — the D-part of the discriminant form is anisotropic (Observation 5.2), and this is verified for L_- itself in `km0places.m`, where the nonzero isotropic cosets of $L_-^\vee/L_-$ number $4, 2, 8$ at $N = 3, 2, 5$, that is $2N-2$. Hence every local factor other than the one at N agrees for φ_μ and φ_0, and with $P(s) = \prod_p W_{0,p}(s,\varphi_0)$,

$$\prod_p W_{0,p}(s,\varphi_\mu) \;=\; \frac{P(s)}{W_{0,N}(s,\varphi_0)},$$

using $W_{0,N}(s,\varphi_\mu) = 1$ from Lemma 11.1. Individual factors have poles at $s=0$ and only the product converges, so this ratio, and not any single place, is the object to evaluate. Writing $A(s)$ for the archimedean factor displayed above, with $A(0) = i\pi$, the vanishing of the $m=0$ coefficient at φ_0 — which is the functional equation, not incoherence — reads $1 = A(0)P(0)$. At $\mu \neq 0$ the term $v^{s/2}\varphi_\mu(0)$ is absent, and by Lemma 11.1 the denominator has a simple pole, so the ratio has a simple zero: $\prod_p W_{0,p}(s,\varphi_\mu) = P(0)\,s\log N/(N-1) + O(s^2)$. Therefore $\kappa_\mu(0) = -A(0)P(0)\log N/(N-1) = -\log N/(N-1)$, every transcendental constant cancelling. Finally the $2N-2$ cosets carry a common $c_\eta(0)$ (Observation 5.2), so the prefactor $-\frac{1}{4}$ of §2 gives $-\frac{1}{4}(2N-2)c_\eta(0)\big(-\log N/(N-1)\big) = \frac{1}{2}c_\eta(0)\log N$. □

Two things are worth drawing out. The factor $\frac{1}{2}$ is $\frac{1}{4}\cdot\frac{2N-2}{N-1}$, hence *independent of N* — which is why the same $\frac{1}{2}$ was measured at $N = 2$, 3 and 5 and why Observation 5.3 could be stated without an N-dependent constant. And the $\log N$ comes from the level prime, not from a place of Diff: not because the level factor *vanishes* — it does not, being identically 1 at every nonzero isotropic coset — but because at the zero coset it has a *pole*, so that replacing it by 1 converts that pole into the zero whose derivative is taken. This is why searching for a vanishing place does not find it, and it is the reason the correction eluded the case analysis of §4, which tracks orders of vanishing only.

Theorem 3.1 and Proposition 4.1 stand in any case: the level factor at $m > 0$ only ever raises the order of vanishing, so κ cannot supply the $\log N$ from the $m > 0$ terms, which is what leaves the whole correction in the $m = 0$ term at $\eta \neq 0$. Part (ii) of the Kudla–Rapoport–Yang programme — the incorporation of a level structure into the moduli problem — remains to be carried out. And what a corrected formula must reproduce is the multipliers of Observation 5.3, equivalently the coefficients of Proposition 10.5; not Table 1 index by index, which fixes the functional only modulo the relation ideal of the panel (Remark 10.6).

A further reading should be recorded, since it would undercut the rest: that the evaluation of $\kappa_\mu^-(m)$ for $m > 0$ is itself wrong at $N > 1$. It had never been checked against an independent computation there — neither here nor, as far as we can determine, in the literature. Yang's worked examples [13, Ex. 21–24] take $B = \left(\frac{-1,3}{\mathbb{Q}}\right)$ with a *maximal* order, and indeed $L = L_+ + L_-$ there; the regression tests in the accompanying code likewise cover only $p = 2$, hyperbolic blocks, and scalings 0 and 1.

That check has now been made where the N-dependence lives. Against the naive local representation density $\alpha_p(m) = \lim_k p^{-k}\#\{x \in (\mathbb{Z}/p^k)^2 : Q(x) \equiv m\}$, computed by counting with convergence in k verified rather than assumed, the code's local Whittaker agrees exactly — with no normalising constant — first in the already-trusted configuration ($p = 2$, scalings 0 and 1), and then in the one that had never been touched: odd $p = 3, 5, 7$ with the lattice p-scaled, i.e. $\mathrm{ord}_p(\det) = 2$, which is the shape of L_- at a firing discriminant, in both the isotropic case (relevant, since firing forces N split) and the anisotropic one. All twelve configurations agree. The closed form $W_{m,N}(1) = (N-1)\,\mathrm{ord}_N(m)$, previously read off the same code that computes it, is likewise reproduced from the counting alone.

This bounds the reading rather than eliminating it: it verifies the local *values*, not the derivative $W'(1)$. But among the $m > 0$ terms the derivative is only ever taken at a vanishing place, and the level prime is never that place in a contributing term — when $N \mid m$ the level factor is $(N-1)\,\mathrm{ord}_N(m) \neq 0$, and when $N \nmid m$ the order of vanishing is at least two and the term is discarded — so there the derivative is always evaluated in a configuration that already occurs at $N = 1$. This is a statement about $m > 0$ only, and the contrast with $m = 0$ is the point of Proposition 11.2: there the level prime *is* where the derivative is taken, and it gets there through a pole rather than a zero, which is exactly the configuration the $m > 0$ analysis never meets.

Remark 11.3 (Published CM values need not discriminate)**.** A natural-seeming check is to compare against a published CM value on a base with $N > 1$ and a nonvanishing correction; $X_0^{14}(5)$ with $s(\tau_{-280}) = 5/16$ [4, Ex. 36] appears to be the only available instance. It cannot work. The Hauptmodul there is normalised by $s(\tau_{-4}) = \infty$, $s(\tau_{-11}) = 1$, $s(\tau_{-35}) = 0$, so the published value depends only on the ratio $s(-280)/s(-35)$ — and -35 and -280 are both non-firing ($5 \mid d$), so any correction multiplies both by the same power of N and cancels; the remaining anchors sit at 0 and ∞, where a multiplicative factor is invisible. Indeed the pipeline reproduces 5/16 exactly both with and without the correction. A published value can only test the correction if its normalising anchors and the tested point lie on opposite sides of the firing rule; this should be checked, at a cost of nothing, before any such comparison is attempted. The consistency relation (2), by contrast, is sensitive exactly where [4, Ex. 36] is blind, and is what supplied the $X_0^{14}(5)$ ground truth here.

Appendix A. Reproducibility

The computations were carried out in Magma against the codebase accompanying this note.

- `VectorValuedForm.m` constructs F_f: coset representatives for $\Gamma_0(M)\backslash SL_2(\mathbb{Z})$, ST-words, the weight 1/2 metaplectic slash, the Weil representation over $\mathbb{C}$, and the Fourier extraction. The action of $\rho(S)$ is implemented as a finite Fourier transform on $L^\vee/L$, which reduces the cost from $|G|^2$ to $|G|\sum_r d_r$ and the memory from $|G|^2$ to $|G|$; without this the case $|G| = 24200$ is out of reach.
- `EisensteinLocalFactors.m` implements [6, (2.16),(2.17), Props. 5.3, 5.4] and exposes `LocalWhittakerPolynomial`, the Whittaker polynomial whose s-derivative produces the logarithms.
- Numerical parameters matter. The horocycle height must satisfy $\Im\tau \geq 1$, or some coset is carried into the pole at ∞ and the terms overwhelm their sum. The number of sample points must beat the positive Fourier coefficients, which grow like $\exp(4\pi\sqrt{pm})$ for pole order p: on the pole-order-30 forms of $X_0^{15}(2)$ the principal-part gate reads $5.8 \cdot 10^{73}$, $1.2 \cdot 10^{62}$, $2.7 \cdot 10^{28}$ and $8.7 \cdot 10^{-103}$ at $K = 48$, 64, 96 and 192 respectively. Raising the working precision without raising K has no effect.
- Every reported value is gated on the principal-part check; a run failing that gate is discarded.

- The exact constant-term evaluation of §8 (`cusp1.m`, `cusp2.m`, and the two-channel solver `ledger2.py`) and the complete campaign data — principal-part dumps, ground-truth sweeps, and the constraints ledger — are banked on the branch `m0-theta-campaign` of the accompanying repository.
- The campaign of §9 is banked on the same branch under `vvdata/weyl-campaign`: the residue-pairing solver (`eis32.m`), the exact rationalisers (`kernelrat.py`, `eisrat.py`), the genus-average drivers (`allgross.m`, `genallgross.py`), the dictionary and mass probes (`grossid100.m`, `grossid900.m`, `swapprobe.m`, `massprobe.m`, `genusfit15sw.m`), the per-base evidence records (`mono`/`epool`/`eis32e`/`kernelrat`/`eisrat`/`allgross` per base), and the composite-s preregistration (`thetasum-form.md`, `mass210.log`).
- The closed form of Proposition 10.5 is implemented independently of the Magma pipeline in `closedcoef.py`, with the Hurwitz class number counted from reduced forms rather than tabulated; it self-checks against the genus-combination coefficients of `ctwm.log` at $D = 15, 39, 55$. `multcheck.py` runs the chain out to the measured multipliers (39 of 39, Remark 10.6) and `gauge152.py` exhibits the rank deficiency that separates it from Table 1.
- For §11: `km0poly.m` computes the level-prime factor at $m = 0$ as a *polynomial* rather than a value, giving the constant 1 of Lemma 11.1; `km0places.m` exhibits the nonzero isotropic cosets of $L_-^\vee/L_-$ at firing discriminants, $2N-2$ of them, supported at the level prime alone; and `km0whit.py` carries out the calibration of Lemma 11.1 against brute-force counts at $p = 3, 5, 7$ and assembles Proposition 11.2. The distinction between a local factor's value and its polynomial matters twice over in this section, and both scripts report the polynomial.
- The bases with $M \geq 660$ are reachable only through `eis32k.m`, which replaces the numerical Weil-representation loop of `eis32.m` by the closed form of [7, Thm. 6.4] (agreement with the numerical driver checked end-to-end at $X_0^{210}(1)$: 960 of 960 words, identical solve residual $1.0087 \cdot 10^{-42}$). Its `EST` switch selects the tracked isotropic coset and is what produces the $\eta = 0$ rows of Remark 9.1; the corresponding control runs reproduce the banked η^* fits exactly. Its `CTL` switch recomputes every ρ-entry by the numerical route and reports the worst deviation; this is the gate through which the odd-level 2-part of Observation 9.8 was admitted (worst $1.3 \cdot 10^{-79}$ over all 144 words of $X_0^{15}(1)$, against $7.3 \cdot 10^{-80}$ for the unchanged even-level path at $X_0^{15}(2)$), run before any fit. Restricted-support gauge checks (the rank exclusions and the degeneracy quoted above) are generated by `genrestrict.py`.

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